

Dual Integral Quadratic Constraints for Robust Control of Partial Integral Equations

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Abstract

Keywords: PDEs, Robust Control, LMIs

1. Introduction

Partial differential equations describe spatially distributed processes whose dynamics cannot generally be captured by a finite-dimensional state without approximation. Relevant applications include thermo-fluidic processes, such as the digital-twin model studied by Toolhally et al. [1]. For such systems, controller synthesis must account for distributed states, boundary conditions, and model uncertainty while retaining guarantees for the original infinite-dimensional system.

The system-theoretic foundations for infinite-dimensional control include the semigroup treatments of Curtain and Pritchard [2] and Curtain and Zwart [3], together with the well-posed linear-systems framework of Staffans [4]. Early robust control results include stabilization by finite-dimensional controllers due to Curtain and Glover [5], and the state-space treatment of H_∞ control for distributed-parameter systems by van Keulen [6].

Several methods address analysis and control directly at the PDE level. Krstić and Smyshlyaev [7] developed backstepping methods for boundary control. Fridman and Blighovsky [8] studied robust Lyapunov-based control, while Valmorbida et al. [9] formulated semidefinite stability tests for a class of PDEs. The PIE representation developed by Peet [10] and extended to input-output systems by Shivakumar et al. [11] provides a bounded-operator state-space framework for PDE analysis. Shivakumar, Das, and Peet [12] subsequently used PIE duality to obtain convex nominal state-feedback synthesis conditions.

Uncertainty is described here using integral quadratic constraints. The IQC framework was introduced by Megretski and Rantzer [13], and its connection with storage inequalities follows the dissipativity theory of Willems [14] and the finite-horizon formulation of

Seiler [15]. Within the PIE framework, Das et al. [16] developed robust analysis for uncertain ODE–PDE systems, Talitckii, Peet, and Seiler [17] established hard IQCs with infinite-dimensional channels, and Lenssen et al. [18] developed structured robustness, performance, and observer-synthesis conditions.

Dual IQCs have previously supported robust feedforward design in the work of Köse and Scherer [19], and estimator and feedforward synthesis for LPV systems in the work of Venkataraman and Seiler [20]. Scherer [21] characterized system structures for which robust synthesis is convex. For more general dynamic multipliers, Veenman and Scherer [22] and Wang, Pfifer, and Seiler [23] used alternating controller and multiplier searches. Yuan and Wu [24] also considered dynamic-IQC feedback for uncertain systems with time-varying state delay.

Finite-horizon use of dynamic multipliers requires particular care in the factorization and storage terms. Carrasco and Seiler [25] studied doubly-hard factorizations, Scherer and Veenman [26] introduced IQCs with terminal costs, and Pfifer and Seiler [27] and Hu et al. [28] showed how multiplier storage can reduce conservatism.

These developments provide the ingredients for robust analysis of PIE systems, but the primal controller-synthesis condition remains nonconvex because the controller couples with both the storage operator and the multiplier. This paper develops an equivalent dual formulation for dynamically filtered IQCs. The dual representation preserves the robust-performance certificate while allowing the state-feedback controller to be synthesized through a convex LPI. It also gives an explicit realization of the resulting controller and relates the operator construction to finite-dimensional dynamic multipliers.

2. Notation & Preliminaries

2.1. Notation

$L_2^n[a, b]$ denotes the set of $\mathbb{R}^n$ -valued, Lebesgue square-integrable equivalence classes of functions on the spatial

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¹*AI use statement.* Generative AI was used as a writing aid in the preparation of this thesis, with all content critically reviewed and verified by the author.

domain $[a, b] \subset \mathbb{R}$. The space $\mathbb{R}L_2^n[a, b]$ denotes the Cartesian product $\mathbb{R}^{n_1} \times L_2^{n_2}[a, b]$, where $\mathbf{n} = \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$, with inner product

$$\left\langle \begin{bmatrix} x_1 \\ \mathbf{x}_2 \end{bmatrix}, \begin{bmatrix} y_1 \\ \mathbf{y}_2 \end{bmatrix} \right\rangle_{\mathbb{R}L_2} := x_1^\top y_1 + \langle \mathbf{x}_2, \mathbf{y}_2 \rangle_{L_2}.$$

When clear from context, we occasionally omit the subscript and simply write $\langle \cdot, \cdot \rangle := \langle \cdot, \cdot \rangle_{\mathbb{R}L_2}$. Since the only spatial domain considered here is $[a, b]$, we typically omit the domain and write L_2^n or $\mathbb{R}L_2^n$, further omitting dimensions when clear from context. We omit dimensions when clear from context. For suitably differentiable spatial states, $\dot{\mathbf{x}} := \partial_t \mathbf{x}$. We denote the induced norm for bounded linear operators on $\mathbb{R}L_2$ by

$$\|\mathcal{P}\| := \sup_{0 \neq \mathbf{x} \in \mathbb{R}L_2^n} \frac{\|\mathcal{P}\mathbf{x}\|_{\mathbb{R}L_2}}{\|\mathbf{x}\|_{\mathbb{R}L_2}}.$$

Our notational convention is to write spatial states and profiles in bold (e.g., $\mathbf{x} \in \mathbb{R}L_2$). For an operator $\mathcal{A}$, the adjoint $\mathcal{A}^*$ is defined by

$$\langle \mathbf{x}, \mathcal{A}\mathbf{y} \rangle_{\mathbb{R}L_2} = \langle \mathcal{A}^*\mathbf{x}, \mathbf{y} \rangle_{\mathbb{R}L_2}, \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}L_2.$$

Define the signal space

$$L_{[a,b]}^{\mathbf{n}} := \{\mathbf{u} : [0, \infty) \rightarrow \mathbb{R}L_2^n[a, b] \mid \|\mathbf{u}\|_{L_{[a,b]}^{\mathbf{n}}}^2 := \int_0^\infty \|\mathbf{u}(t)\|_{\mathbb{R}L_2}^2 dt < \infty\},$$

with inner product $\langle \mathbf{u}, \mathbf{v} \rangle_{L_{[a,b]}^{\mathbf{n}}} := \int_0^\infty \langle \mathbf{u}(t), \mathbf{v}(t) \rangle_{\mathbb{R}L_2} dt$. For any $\mathbf{y} : [0, \infty) \rightarrow \mathbb{R}L_2^n[a, b]$, define the truncation operator $(p_T \mathbf{y})(t) := \mathbf{y}(t)$ for $0 \leq t \leq T$ and $(p_T \mathbf{y})(t) := 0$ for $t > T$, the corresponding extended signal space is therefore,

$$L_{e,[a,b]}^{\mathbf{n}} := \{\mathbf{v} \mid p_T \mathbf{v} \in L_{[a,b]}^{\mathbf{n}} \forall T > 0\}.$$

Moreover, for $\tau \in \mathbb{R}$ define the shift operator $(s_\tau u)(t) := u(t - \tau)$. An operator $G : L_{e,[a,b]}^{\mathbf{n}} \rightarrow L_{e,[a,b]}^{\mathbf{m}}$ is called

- *Causal*, if $p_T G = p_T G p_T$ for all $T > 0$.
- *Anticausal*, if $(I - p_T)G(I - p_T) = (I - p_T)G$ for all $T > 0$.
- *Memoryless*, if G is simultaneously causal and anticausal.
- *Time-Invariant*, if $s_\tau G = G s_\tau$ for all $\tau \in \mathbb{R}$.
- *Bounded on $L_{[a,b]}$* , if $\|G\| \leq C$ for some $C \geq 0$.
- *Bounded on $L_{e,[a,b]}$* , if $\|p_T G\| \leq C$ for some $C \geq 0$ and all $T > 0$.

For a bounded linear operator $\mathcal{K} : \mathbb{R}L_2^n \rightarrow \mathbb{R}L_2^m$, define the multiplication operator $\mathcal{M}_{\mathcal{K}} : L_{e,[a,b]}^{\mathbf{n}} \rightarrow L_{e,[a,b]}^{\mathbf{m}}$ by

$$(\mathcal{M}_{\mathcal{K}} u)(t) := \mathcal{K}u(t).$$

2.2. Partial Integral Equations (PIEs)

Partial Integral operators, also known as PI operators, are integral operators defined jointly on a vector space and Hilbert space [10, 11].

Definition 1. We say an operator $\mathcal{P} : \mathbb{R}L_2^n[a, b] \rightarrow \mathbb{R}L_2^m[a, b]$ belongs to $\Pi_4^{m \times n}$, denoted by $\Pi \left[\begin{array}{c|c} P & Q_1 \\ \hline Q_2 & \{R_0, R_1, R_2\} \end{array} \right]$, if there exist a matrix P and matrix polynomials Q_1, Q_2, R_0, R_1 , and R_2 (of compatible dimension) such that,

$$\left(\left(\Pi \left[\begin{array}{c|c} P & Q_1 \\ \hline Q_2 & \{R\} \end{array} \right] \right) \begin{bmatrix} x \\ \mathbf{x} \end{bmatrix} \right) (s) := \begin{bmatrix} Px + \int_a^b Q_1(s) \mathbf{x}(s) ds \\ Q_2(s)x + \mathcal{R}\mathbf{x}(s) \end{bmatrix},$$

$$(\mathcal{R}\mathbf{x})(s) = R_0(s)\mathbf{x}(s) + \int_a^s R_1(s, \theta)\mathbf{x}(\theta) d\theta + \int_s^b R_2(s, \theta)\mathbf{x}(\theta) d\theta.$$

The vector space of PI operators, given compatible dimensions, are closed under composition, addition, adjoint and concatenation – implying that Π_4 is a composition algebra [11].

Definition 2. (LPIs) For given PI operators $\{\mathcal{E}_{i,j}, \mathcal{F}_{i,j}, \mathcal{G}_i\}$ and a convex linear functional $\ell(\cdot)$, a linear PI inequality is a convex optimization of the following form

$$\min_{P_i, Q_{1i}, Q_{2i}, R_{0i}, R_{1i}, R_{2i}} \ell(\{P_i, Q_{1i}, Q_{2i}, R_{0i}, R_{1i}, R_{2i}\})$$

$$\text{s.t. } \sum_{j=1}^K \mathcal{E}_{ij}^* \Pi \left[\begin{array}{c|c} P_i & Q_{1i} \\ \hline Q_{2i} & \{R_i\} \end{array} \right] \mathcal{F}_{ij} + \mathcal{G}_i \succeq 0$$

Only LMIs are required to solve an optimization problem involving LPIs. We next describe PIE systems and their stability properties, following [12]. Given the PIE system $G\{\mathcal{T}, \mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}\}$ with PI operators $\mathcal{T}, \mathcal{A}, \mathcal{B}, \mathcal{C}$, and $\mathcal{D}$ of compatible dimensions, and signals $w(t) \in \mathbb{R}L_2^{n_w}[a, b]$, we say $\mathbf{x}(t) \in \mathbb{R}L_2^{n_x}[a, b]$ and $z(t) \in \mathbb{R}L_2^{n_z}[a, b]$ satisfy the PIE with initial condition $\mathbf{x}_0$ if

$$\begin{bmatrix} \partial_t(\mathcal{T}\mathbf{x})(t) \\ z(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ w(t) \end{bmatrix}$$

for all $t \geq 0$ and $\mathcal{T}\mathbf{x}(0) = \mathcal{T}\mathbf{x}_0$. Additionally, for signals $\underline{w}(t) \in \mathbb{R}L_2^{n_z}[a, b]$ of compatible dimensions, we say $\underline{\mathbf{x}}(t) \in \mathbb{R}L_2^{n_x}[a, b]$ and $\underline{z}(t) \in \mathbb{R}L_2^{n_w}[a, b]$ satisfy the dual PIE, $G^\top$, with initial condition $\underline{\mathbf{x}}_0$ if

$$\begin{bmatrix} \mathcal{T}^* \underline{\dot{\mathbf{x}}}(t) \\ \underline{z}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}^* & \mathcal{C}^* \\ \mathcal{B}^* & \mathcal{D}^* \end{bmatrix} \begin{bmatrix} \underline{\mathbf{x}}(t) \\ \underline{w}(t) \end{bmatrix},$$

for all $t \geq 0$ and $\mathcal{T}^* \underline{\mathbf{x}}(0) = \mathcal{T}^* \underline{\mathbf{x}}_0$. For these systems, we consider input–output and exponential stability. A notational convention is using underlining anything related to the dual system. Given a PIE system G , we say G is input–output stable if for any $z, w \in L_{e,[a,b]}$ satisfying $z = Gw$ with $\mathbf{x}_0 = 0$, we have

$$\|p_T z\|_{L_{[a,b]}} \leq \gamma \|p_T w\|_{L_{[a,b]}}, \quad \forall T \geq 0, \quad \forall w \in L_{e,[a,b]}^{n_w}.$$

Given a PIE system G , we say that the homogeneous part of G is *exponentially stable* with decay rate $\alpha > 0$ if there exists some $M > 0$ such that for any $\mathbf{x}_0 \in \mathbb{R}L_2$, if $\mathcal{T}\mathbf{x}(0) = \mathcal{T}\mathbf{x}_0$ and $\mathcal{T}\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t)$, then

$$\|\mathcal{T}\mathbf{x}(t)\|_{\mathbb{R}L_2} \leq M\|\mathbf{x}_0\|_{\mathbb{R}L_2}e^{-\alpha t} \quad \forall t \geq 0.$$

To describe feedback interconnections, we use the following notation. For compatible causal G and H , the Redheffer star products [29] denote the feedback maps

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = G \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \quad \begin{array}{ll} G \star H : u_2 \mapsto y_2 & u_1 = Hy_1 \\ H \star G : u_1 \mapsto y_1 & u_2 = Hy_2 \end{array},$$

provided the respective interconnections are well-posed. Here, $P \star \Delta$ closes the uncertainty loop for the generalized plant P , while $K \star G$ closes the controller loop for the controlled plant G .

2.3. PIETOOLS

PIETOOLS is a MATLAB toolbox developed by Shivakumar, Das, and Peet [30] for declaring, manipulating, and solving LPIs. Its operator syntax, PDE-to-PIE conversion, and optimization workflows are documented by Shivakumar et al. in the user manual [31]. See also the PIETOOLS homepage.

3. IQC Theory

This section introduces the IQC objects used to study stability and performance of interconnected systems in a Hilbert space. The construction builds on the IQC framework of Megretski and Rantzer [13] and its extension to hard IQCs for infinite-dimensional channels [17]. We establish the generalized system interconnection, formalize stability and performance, and define the uncertainty and performance IQCs, as described in [18]. These definitions provide the foundation for the theory developed in sections 4, 5, and 6.

3.1. Description of uncertain systems in a Hilbert-space

We consider operators that map external inputs to external outputs through a feedback interconnection between a linear (infinite dimensional) operators P and Δ . This interconnection structure is formally defined as follows

Definition 3 (Generalized Plant). *Given operators $P_\Delta : L_{e,[a,b]}^{\mathbf{n}_{w\Delta}} \rightarrow L_{e,[a,b]}^{\mathbf{n}_{z\Delta}}$, $P_{\Delta P} : L_{e,[a,b]}^{\mathbf{n}_{wp}} \rightarrow L_{e,[a,b]}^{\mathbf{n}_{z\Delta}}$, $P_{P\Delta} : L_{e,[a,b]}^{\mathbf{n}_{w\Delta}} \rightarrow L_{e,[a,b]}^{\mathbf{n}_{zp}}$, $P_P : L_{e,[a,b]}^{\mathbf{n}_{wp}} \rightarrow L_{e,[a,b]}^{\mathbf{n}_{zp}}$, and $\Delta : L_{e,[a,b]}^{\mathbf{n}_{z\Delta}} \rightarrow L_{e,[a,b]}^{\mathbf{n}_{w\Delta}}$, define*

$$\begin{bmatrix} z_\Delta \\ z_P \end{bmatrix} = P \begin{bmatrix} w_\Delta \\ w_P \end{bmatrix} = \begin{bmatrix} P_\Delta & P_{\Delta P} \\ P_{P\Delta} & P_P \end{bmatrix} \begin{bmatrix} w_\Delta \\ w_P \end{bmatrix}, \quad w_\Delta = \Delta z_\Delta. \quad (1)$$

for input $w_P \in L_{e,[a,b]}^{\mathbf{n}_{wp}}$ and output $z_P \in L_{e,[a,b]}^{\mathbf{n}_{zp}}$, there exist internal signals $z_\Delta \in L_{e,[a,b]}^{\mathbf{n}_{z\Delta}}$ and $w_\Delta \in L_{e,[a,b]}^{\mathbf{n}_{w\Delta}}$ satisfying the interconnected system $z_P = (P \star \Delta)w_P$.

The interconnection structure described above captures the feedback interaction between the nominal system P and the uncertainty Δ , where w_P and z_P represent external inputs and performance outputs, respectively, while w_Δ and z_Δ form the internal feedback loop through the uncertainty. Before analyzing stability and performance, we must first ensure the well-posedness of the interconnection, such that solutions exist, are unique, and depend causally on inputs.

Definition 4. *We say that the uncertain interconnection $P \star \Delta$ is well-posed if*

- Existence and Uniqueness, *For any $w \in L_{e,[a,b]}$, there exist unique $z \in L_{e,[a,b]}$, $z_\Delta \in L_{e,[a,b]}$ and $w_\Delta \in L_{e,[a,b]}$ such that, z , w , z_Δ , w_Δ satisfy the interconnection.*
- Causality, *If z_Δ , w_Δ and $\hat{z}_\Delta$, $\hat{w}_\Delta$ satisfy the interconnection, then $p_T(z_\Delta - \hat{z}_\Delta) = 0$ and $p_T(w_\Delta - \hat{w}_\Delta) = 0$, for all $T \geq 0$.*

Having established the class of uncertain systems and their well-posedness, we now develop a framework to guarantee robust stability and performance for the class of systems defined in Definition 3.

3.2. Robust stability and performance

Veenman, Scherer, and K ro glu [32] describe both uncertainty and performance specifications through quadratic constraints on interconnection signals. For PIE systems, this formulation also supports structured robustness and performance analysis [18]. We now define what it means for the system to remain stable despite uncertainties. Robust stability requires that the system is well-posed for all admissible uncertainties and that the input-output map remains bounded. This involves removal of the performance channels in (1), and introducing auxiliary signals z_0 and w_0 .

Definition 5. *The interconnection*

$$z_\Delta = P_\Delta w_\Delta + z_0, \quad w_\Delta = \Delta z_\Delta + w_0,$$

with $z_0, w_0 \in L_{e,[a,b]}$, is said to be robustly stable if

1. *The uncertain system is well-posed for all $\Delta \in \Delta$.*
2. *The map from $\begin{bmatrix} z_0 \\ w_0 \end{bmatrix}$ to $\begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix}$ is bounded on $L_{e,[a,b]}$ for all $\Delta \in \Delta$.*

Within the IQC framework, the uncertainty operator Δ is described by a quadratic constraint defined by an appropriate multiplier. A stability certificate combines this constraint with a complementary condition on the plant. We use the finite-horizon, or hard, IQC formulation [15, 17], leading to the following definition.

Definition 6. *An uncertainty operator $\Delta : L_{e,[a,b]}^{\mathbf{n}_{z\Delta}} \rightarrow L_{e,[a,b]}^{\mathbf{n}_{w\Delta}}$ is said to satisfy the hard IQC $(\Psi_\Delta, \mathcal{M}_{\mathcal{V}_\Delta})$ defined*

$$\begin{aligned}
G &:= \begin{cases} \begin{bmatrix} \mathcal{T}_G \dot{\mathbf{x}}_G(t) \\ z(t) \\ y_G(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_G & \mathcal{B}_G & \mathcal{B}_u \\ \mathcal{C}_G & \mathcal{D}_G & \mathcal{D}_{zu} \\ I & 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_G(t) \\ w(t) \\ u(t) \end{bmatrix} \\ K &:= \begin{cases} u(t) = [\mathcal{K}_G \quad \mathcal{K}_\Theta] \begin{bmatrix} \mathbf{x}_G(t) \\ \mathbf{x}_\Theta(t) \end{bmatrix} \end{cases} \\ \Theta &:= \begin{cases} \begin{bmatrix} \mathcal{T} \dot{\mathbf{x}}_\Theta(t) \\ \tilde{v}(t) \\ y_\Theta(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \\ I & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_\Theta(t) \\ v(t) \end{bmatrix} \end{cases} \\ \Theta \begin{bmatrix} K \star G \\ I \end{bmatrix} &:= \begin{cases} \begin{bmatrix} \mathcal{T}_G \dot{\mathbf{x}}_G(t) \\ \mathcal{T} \dot{\mathbf{x}}_\Theta(t) \\ \tilde{z}(t) \\ \tilde{w}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_G + \mathcal{B}_u \mathcal{K}_G & \mathcal{B}_u \mathcal{K}_\Theta & \mathcal{B}_G \\ \mathcal{B}_1(\mathcal{C}_G + \mathcal{D}_{zu} \mathcal{K}_G) & \mathcal{A} + \mathcal{B}_1 \mathcal{D}_{zu} \mathcal{K}_\Theta & \mathcal{B}_1 \mathcal{D}_G + \mathcal{B}_2 \\ \mathcal{D}_{11}(\mathcal{C}_G + \mathcal{D}_{zu} \mathcal{K}_G) & \mathcal{C}_1 + \mathcal{D}_{11} \mathcal{D}_{zu} \mathcal{K}_\Theta & \mathcal{D}_{11} \mathcal{D}_G + \mathcal{D}_{12} \\ \mathcal{D}_{21}(\mathcal{C}_G + \mathcal{D}_{zu} \mathcal{K}_G) & \mathcal{C}_2 + \mathcal{D}_{21} \mathcal{D}_{zu} \mathcal{K}_\Theta & \mathcal{D}_{21} \mathcal{D}_G + \mathcal{D}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{x}_G(t) \\ \mathbf{x}_\Theta(t) \\ w(t) \end{bmatrix} \end{cases} \end{cases} \quad (2)
\end{aligned}$$

Figure 1: Primal component realizations and full augmented interconnection for state feedback with dynamic filtering.

by the dynamic operator $\Psi_\Delta : L_{e,[a,b]}^{\mathbf{n}_{z_\Delta} + \mathbf{n}_{w_\Delta}} \rightarrow L_{e,[a,b]}^{\mathbf{n}_{z_\Delta} + \mathbf{n}_{w_\Delta}}$ and the bounded self-adjoint operator $\mathcal{V}_\Delta = \mathcal{V}_\Delta^*$ if the inequality holds for all $z_\Delta \in L_{e,[a,b]}$ and all $T \geq 0$,

$$\left\langle p_T \Psi_\Delta \begin{bmatrix} z_\Delta \\ \Delta z_\Delta \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_\Delta} \Psi_\Delta \begin{bmatrix} z_\Delta \\ \Delta z_\Delta \end{bmatrix} \right\rangle_{L_{[a,b]}} \geq 0.$$

Moreover, let $\mathcal{V}_\Delta \{\mathcal{Q}_\Delta, \mathcal{S}_\Delta, \mathcal{R}_\Delta\}$ be parameterized as,

$$\mathcal{V}_\Delta = \begin{bmatrix} \mathcal{Q}_\Delta & \mathcal{S}_\Delta \\ \mathcal{S}_\Delta^* & \mathcal{R}_\Delta \end{bmatrix}$$

Beyond stability, we are interested in ensuring that the system meets certain performance specifications in the presence of uncertainty. Robust performance extends the notion of robust stability by additionally requiring that a performance metric is satisfied.

Definition 7. Robust performance is achieved if the interconnection $P \star \Delta$ is robustly stable and for performance IQC $(\Psi_p, \mathcal{M}_{\mathcal{V}_p})$ defined by $\Psi_p : L_{e,[a,b]}^{\mathbf{n}_{z_p} + \mathbf{n}_{w_p}} \rightarrow L_{e,[a,b]}^{\mathbf{n}_{z_p} + \mathbf{n}_{w_p}}$ and $\mathcal{V}_p = \mathcal{V}_p^*$, such that

$$\left\langle p_T \Psi_p \begin{bmatrix} v \\ 0 \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_p} \Psi_p \begin{bmatrix} v \\ 0 \end{bmatrix} \right\rangle_{L_{[a,b]}} \geq \epsilon$$

for any $v \in L_{e,[a,b]}^{\mathbf{n}_{z_p}}$ we have

$$\left\langle p_T \Psi_p \begin{bmatrix} P \star \Delta \\ I \end{bmatrix} w_p, p_T \mathcal{M}_{\mathcal{V}_p} \Psi_p \begin{bmatrix} P \star \Delta \\ I \end{bmatrix} w_p \right\rangle_{L_{[a,b]}} \leq -\epsilon \|p_T w_p\|_{L_{[a,b]}}^2.$$

for all $w_p \in L_{e,[a,b]}^{\mathbf{n}_{w_p}}$ and $T \geq 0$.

In this framework, the multipliers $\mathcal{M}_{\mathcal{V}_\Delta}$ and $\mathcal{M}_{\mathcal{V}_p}$ serve distinct but complementary roles. The hard uncertainty IQC $(\Psi_\Delta, \mathcal{M}_{\mathcal{V}_\Delta})$ characterizes the behavior of uncertainties, nonlinearities, and other perturbations captured by the operator Δ , ensuring that the IQC is satisfied for the class of admissible disturbances. In contrast, the performance IQC $(\Psi_p, \mathcal{M}_{\mathcal{V}_p})$ encodes performance requirements

for the nominal system, such as L_2 -gain, passivity conditions, or other desired input-output properties. Together, these multipliers enable simultaneous verification of robust stability against perturbations and achievement of specified performance objectives. This section applies the IQC framework to the primal system and gives a certificate for simultaneous robust stability and performance. For brevity, we write the hard IQC as

$$\left\langle p_T \Psi \begin{bmatrix} Pw \\ w \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}} \Psi \begin{bmatrix} Pw \\ w \end{bmatrix} \right\rangle_{L_{[a,b]}} \leq -\epsilon \|p_T w\|_{L_{[a,b]}}^2.$$

Let Ψ_Δ and Ψ_p have a 2×2 block partition and define $\Psi_{ij} := \text{diag}((\Psi_\Delta)_{ij}, (\Psi_p)_{ij})$ for $i, j \in \{1, 2\}$. Let $w^\top = [w_\Delta^\top \quad w_p^\top]$ and $z^\top = [z_\Delta^\top \quad z_p^\top]$. Moreover, define $\mathcal{Q} = \text{diag}(\mathcal{Q}_\Delta, \mathcal{Q}_p)$, $\mathcal{S} = \text{diag}(\mathcal{S}_\Delta, \mathcal{S}_p)$, and $\mathcal{R} = \text{diag}(\mathcal{R}_\Delta, \mathcal{R}_p)$, then $\mathcal{V}\{\mathcal{Q}, \mathcal{S}, \mathcal{R}\}$.

Theorem 1 (Robust Stability and Performance). *Let P, Δ be given, and there exists $\Psi_\Delta, \mathcal{V}_\Delta$, such that,*

- The interconnection $P \star \Delta$ is well-posed for all $\Delta \in \Delta$
- Any uncertainty $\Delta \in \Delta$ satisfy the hard IQC defined by Ψ_Δ and $\mathcal{V}_\Delta$.

Then the interconnection $P \star \Delta$ is robustly stable and robust performance is achieved with respect to some Ψ_p and $\mathcal{V}_p = \{\mathcal{Q}_p, \mathcal{S}_p, \mathcal{R}_p\}$, with $\mathcal{Q}_p \succeq 0$, if,

$$\begin{aligned}
&\left\langle p_T \Psi \begin{bmatrix} Pw \\ w \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}} \Psi \begin{bmatrix} Pw \\ w \end{bmatrix} \right\rangle_{L_{[a,b]}} \\
&\leq -\epsilon \|p_T w\|_{L_{[a,b]}}^2
\end{aligned} \quad (3)$$

for all $w_\Delta \in L_{e,[a,b]}^{\mathbf{n}_{w_\Delta}}$, $w_p \in L_{e,[a,b]}^{\mathbf{n}_{w_p}}$ and $T \geq 0$.

Proof. See Appendix A.1. ■

Theorem 1 establishes a framework for deriving Linear PI Inequalities (LPIs), which are used in assessing the robust performance of a system.

4. Problem Definition (Primal System)

A recent state-space framework based on Partial Integral Equations (PIEs) provides a unified formulation of stabilization and H_∞ -optimal state-feedback controller synthesis for PDEs [12]. This framework addresses the nominal setting, corresponding to $\Delta = 0$. Building on this result, we develop a unified framework for optimal state-feedback control of uncertain or nonlinear PIEs subject to a prescribed quadratic performance requirement. As established in the previous section, IQCs are used to characterize the uncertainty and reduce conservatism in the resulting robust-stability certificates. Let G denote the open-loop PIE plant to which the controller is connected. Its realization is given in Eq. (2). For a controller K , define the closed-controller generalized plant as $P_K := K \star G$.

Our objective is to construct a state-feedback controller K such that $P_K \star \Delta$ is robustly stable and satisfies the prescribed performance IQC for every $\Delta \in \mathbf{\Delta}$. We consider the controller interconnection shown in Fig. 2, with

$$u = \mathcal{K}_G \mathbf{x}_G + \mathcal{K}_\Theta \mathbf{x}_\Theta. \quad (4)$$

The auxiliary filter Θ and its associated state $\mathbf{x}_\Theta$ are constructed below. For dynamic IQCs, causal implementation of the state-feedback law requires that the signals w and z driving the dynamic part of Θ be available to the controller, so that $\mathbf{x}_\Theta$ can be computed in real time.

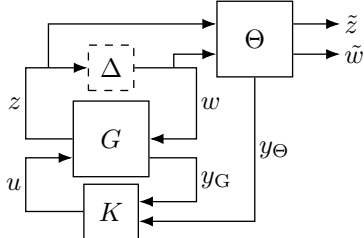


Figure 2: Primal controller interconnection.

4.1. Inputs at the boundary

We also admit boundary control and disturbance inputs through the treatment introduced by Shivakumar et al. [12, Sec. III-B]. The following construction and heat-equation example are adapted directly from that work. Let $x(t, s) = \partial_s^2 v(t, s)$ and impose the boundary conditions $v(t, 0) = u_1(t)$ and $\partial_s v(t, 0) = u_2(t)$. The distributed state is reconstructed as

$$v(t, s) = u_1(t) + su_2(t) + \int_0^s (s - \eta)x(t, \eta) \, d\eta. \quad (5)$$

For the heat equation $\partial_t v = \partial_s^2 v$, this gives

$$\partial_t \int_0^s (s - \eta)x(t, \eta) \, \mathrm{d}\eta = x(t, s) - \dot{u}_1(t) - s\dot{u}_2(t). \quad (6)$$

Thus a boundary signal enters the PIE through its time derivative. Retaining this derivative while measuring the

gain from the undifferentiated signal in L_2 would require additional temporal regularity. Following [12], for a generic boundary signal w_b we therefore set $\hat{w}_b = \dot{w}_b$ and regard $\hat{w}_b$ as the PIE input.

If w_b is required by the controller or an IQC filter, it is recovered through

$$\dot{w}_b(t) = \hat{w}_b(t), \quad w_b(t) = w_b(0) + \int_0^t \hat{w}_b(\tau) d\tau. \quad (7)$$

We assume that $w_b(0)$ is known and that w_b is available wherever it is used. For the zero-state performance problem, $w_b(0) = 0$, consistently with the initial boundary value of the PDE. The integrator state is included in the augmented plant realization. Hence the system class includes distributed inputs and boundary control or disturbance inputs for which this augmentation produces a well-posed PIE realization. Direct uncertainty through a boundary condition is outside this class unless it can also be represented by a well-posed PIE uncertainty channel satisfying the assumed hard IQC.

4.2. Primal analysis and synthesis obstruction

To distinguish analysis from synthesis, fix the IQC filter Ψ and write an augmented realization before closing the controller as

$$\begin{bmatrix} \mathcal{T}\dot{\mathbf{x}} \\ \tilde{v} \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B}_{\text{u}} & \mathcal{B} \\ \mathcal{C} & \mathcal{D}_{\text{zu}} & \mathcal{D} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ u \\ w \end{bmatrix}. \quad (8)$$

With $u = \mathcal{K}\mathbf{x}$, where $\mathcal{K} = [\mathcal{K}_P \quad \mathcal{K}_\Theta]$, the filtered graph $\Psi \begin{bmatrix} K \star G \\ I \end{bmatrix}$ has the PIE realization

$$\{\mathcal{T}, \mathcal{A} + \mathcal{B}_u \mathcal{K}, \mathcal{B}, \mathcal{C} + \mathcal{D}_{zu} \mathcal{K}, \mathcal{D}\}.$$

A sufficient condition for robust stability and performance is the existence of a PI operator $\mathcal{P} = \mathcal{P}^* \succeq 0$ and $\epsilon > 0$ satisfying

$$\begin{aligned} & \begin{bmatrix} \mathcal{T}^* \mathcal{P}(\mathcal{A} + \mathcal{B}_u \mathcal{K}) + (\mathcal{A} + \mathcal{B}_u \mathcal{K})^* \mathcal{P} \mathcal{T} & \mathcal{T}^* \mathcal{P} \mathcal{B} \\ \mathcal{B}^* \mathcal{P} \mathcal{T} & \epsilon I \end{bmatrix} \\ & + \begin{bmatrix} (\mathcal{C} + \mathcal{D}_{zu} \mathcal{K})^* \\ \mathcal{D}^* \end{bmatrix} \mathcal{V} [(\mathcal{C} + \mathcal{D}_{zu} \mathcal{K}) \quad \mathcal{D}] \preceq 0. \end{aligned} \quad (9)$$

For analysis, $\mathcal{K}$ is fixed and (9) is affine in the certificate operators $\mathcal{P}$ and $\mathcal{V}$. For synthesis, these operators and $\mathcal{K}$ must be determined simultaneously. This introduces products between $\mathcal{P}$ and $\mathcal{K}$, and between $\mathcal{K}$ and $\mathcal{V}$. For a non-descriptor ODE, the storage and controller coupling can be removed with $\mathcal{Q} = \mathcal{P}^{-1}$ and $\mathcal{Z} = \mathcal{K}\mathcal{Q}$. The reconstruction operator $\mathcal{T}$ prevents the same cancellation for a general PIE. The multiplier coupling must likewise be resolved by additional structure. Thus the primal formulation provides a convex test for a fixed controller, but not a convex synthesis condition. The following sections construct a dual equivalent of the filtered interconnection from which K can be synthesized convexly.

5. Duality

We consider a generalized plant P with a PIE state-space realization. The nominal PIE duality results of Shiva-kumar et al. [12] provide the starting point for convex state-feedback synthesis. For dynamically filtered IQCs, two further ingredients are relevant, hard factorizations connecting IQS theory to dissipativity theory [15], and the dual-filter and storage transformations developed for LPV estimation and feedforward synthesis [20]. This section develops the corresponding dual-filter construction for PIE systems using the Potapov–Ginzburg transform and PIE gain duality.

5.1. Nominal PIE gain duality

We begin by restating the main theorem from [12]. In [12], the theorem is formulated for the case where $\mathcal{D}$ is a matrix and the induced L_2 norm is bounded over an infinite time horizon. Generalizing this result to the case where $\mathcal{D}$ is a PI operator and considering the bounded truncated induced $L_{[a,b]}$ norm does not alter the proof.

Theorem 2 (Dual L_2 -gain [12, Thm. 9]). *For a PIE system with corresponding PI operators $P\{\mathcal{T}, \mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}\}$, the following statements are equivalent.*

1. For any $w \in L_{e,[a,b]}^{\mathbf{n}_w}$, and $\mathbf{x}(0) = 0$, P satisfies $\|p_T z\| \leq \gamma \|p_T w\|$ for all $T > 0$.
2. For any $\underline{w} \in L_{e,[a,b]}^{\mathbf{n}_z}$, and $\underline{\mathbf{x}}(0) = 0$, $P^\top$ satisfies $\|p_T \underline{z}\| \leq \gamma \|p_T \underline{w}\|$ for all $T > 0$.

For a bounded, time-invariant, memoryless linear uncertainty Δ , suppose the feedback interconnection is well-posed and its closed-loop operators satisfy the hypotheses of [12, Thm. 6]. Closing the loop $w = \Delta z$ gives

$$\mathcal{T}\dot{\mathbf{x}} = (\mathcal{A} + \mathcal{B}\Delta(I - \mathcal{D}\Delta)^{-1}\mathcal{C})\mathbf{x}.$$

Taking adjoints and using the push-through identity yields

$$\begin{aligned} (\mathcal{T})^* &= \mathcal{T}^*, & (\mathcal{A} + \mathcal{B}\Delta(I - \mathcal{D}\Delta)^{-1}\mathcal{C})^* \\ &= \mathcal{A}^* + \mathcal{C}^*\Delta^*(I - \mathcal{D}^*\Delta^*)^{-1}\mathcal{B}^*. \end{aligned}$$

This is exactly the state operator of $P^\top \star \Delta^*$. Thus, [12, Thm. 6] establishes equivalence of weak convergence for the primal and dual states. Exponential stability is likewise equivalent by [12, Cor. 8]. For the constant repeated real uncertainty $\Delta = \delta I$ in Section 7.2, $\Delta^* = \Delta$, so the same uncertainty appears in both interconnections.

5.2. Duality under dynamic filtering

For a general uncertainty Δ , this construction is less trivial. It is therefore more convenient to develop duality through the IQC description, whose filters and multiplier weights are linear operators, rather than through Δ itself. The explicit uncertainty-adjoint relation of [19] assumes

linear uncertainty. Venkataraman and Seiler [20] accommodate nonlinear uncertainties through dual IQCs and J -spectral factorizations of strict positive-negative multipliers. We adopt the same strategy. We first reformulate the IQC condition as a contraction requirement on an associated system G_0 with respect to the $L_{e,[a,b]}$ norm, then apply the previously stated theorem, and finally recover the original formulation using the dual filter. In their framework, this relationship is established through congruence transformations of the analysis LMIs [20, Sec. 3]. Our approach relies on PIE gain duality together with truncated signal inequalities imposed over arbitrary finite horizons. This provides a natural extension of the methodology to robust feedback synthesis for PDE systems.

Definition 8 (J -Spectral Factorization and Dual Filter). *For a given IQC defined by $(\Psi, \mathcal{V})$, we say that a causal linear bounded operator Θ and $J_{\mathbf{n}_z, \mathbf{n}_w} := \begin{bmatrix} I_{\mathbf{n}_z} & 0 \\ 0 & -I_{\mathbf{n}_w} \end{bmatrix}$ (the pair $(\Theta, J_{\mathbf{n}_z, \mathbf{n}_w})$) is a J -spectral factorization of $(\Psi, \mathcal{V})$ if,*

$$\begin{aligned} &\left\langle p_T \Psi \begin{bmatrix} z \\ w \end{bmatrix}, p_T \mathcal{M}_\Psi \Psi \begin{bmatrix} z \\ w \end{bmatrix} \right\rangle_{L_{[a,b]}} \\ &= \left\langle p_T \Theta \begin{bmatrix} z \\ w \end{bmatrix}, p_T J_{\mathbf{n}_z, \mathbf{n}_w} \Theta \begin{bmatrix} z \\ w \end{bmatrix} \right\rangle_{L_{[a,b]}}, \end{aligned}$$

Following the dual-filter construction in [20], if Θ has a causal bounded inverse, we use $\mathbf{D}(\Theta)$ to denote the dual filter associated with Θ of the form

$$\mathbf{D}(\Theta) := (\hat{J}^* \Theta^\top \hat{J})^{-1},$$

$$\text{where } \hat{J} := \begin{bmatrix} 0_{\mathbf{n}_z \times \mathbf{n}_w} & I_{\mathbf{n}_z} \\ -I_{\mathbf{n}_w} & 0_{\mathbf{n}_w \times \mathbf{n}_z} \end{bmatrix}.$$

To state the equivalence compactly, define the primal quadratic form

$$\mathcal{I} := \left\langle p_T \Theta \begin{bmatrix} Pw \\ w \end{bmatrix}, p_T J_{\mathbf{n}_z, \mathbf{n}_w} \Theta \begin{bmatrix} Pw \\ w \end{bmatrix} \right\rangle_{L_{[a,b]}}.$$

The corresponding dual quadratic form is

$$\underline{\mathcal{I}} := \left\langle p_T \mathbf{D}(\Theta) \begin{bmatrix} P^\top \underline{w} \\ \underline{w} \end{bmatrix}, p_T J_{\mathbf{n}_w, \mathbf{n}_z} \mathbf{D}(\Theta) \begin{bmatrix} P^\top \underline{w} \\ \underline{w} \end{bmatrix} \right\rangle_{L_{[a,b]}}.$$

Theorem 3. *Let P be as in Definition 3 and let $(\Theta, J_{\mathbf{n}_z, \mathbf{n}_w})$ be a hard J -spectral factorization. Moreover, let P and Θ admit a PIE realization. Finally, assume $(\Theta_{21}P + \Theta_{22})$ has a bounded causal inverse. Define*

$$G_0 = (\Theta_{11}P + \Theta_{12})(\Theta_{21}P + \Theta_{22})^{-1}$$

then the following statements are equivalent,

1. *There exists some $\rho < 1$ such that $\|p_T G_0 \tilde{w}\|_{L_{[a,b]}}^2 < \rho \|p_T \tilde{w}\|_{L_{[a,b]}}^2$ for every $\tilde{w}$ and $T \geq 0$.*
2. *There exists some $\epsilon > 0$ such that $\mathcal{I} \leq -\epsilon \|p_T w\|_{L_{[a,b]}}^2$ for every w and $T \geq 0$.*
3. *There exists some $\underline{\epsilon} > 0$ such that $\underline{\mathcal{I}} \leq -\underline{\epsilon} \|p_T \underline{w}\|_{L_{[a,b]}}^2$ for every $\underline{w}$ and $T \geq 0$.*

Proof. See Appendix A.2. ■

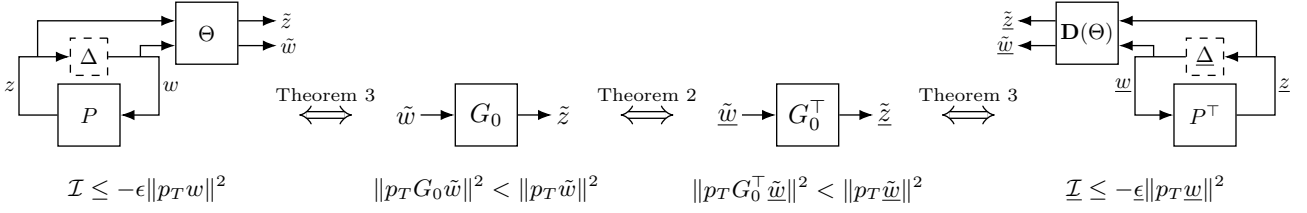


Figure 3: Equivalence chain for primal and dual dynamic filtering.

5.3. Dual robust-performance certificate

Theorem 3 identifies the strict primal and dual IQC inequalities through a common contraction bound on G_0 . Either side of the filtered graph may therefore be used to construct the same performance certificate. The resulting KYP condition permits the robust-performance certificate to be imposed on either the primal or dual realization.

Theorem 4. *Let the realization of P satisfy Definition 3. Let $\Delta \in \mathbf{\Delta}$ and there exist either, a primal hard IQC $(\Psi, \mathcal{M}_\Psi)$, with $\Psi \in \Pi_4$, or a dual hard IQC $(\underline{\Psi}, \mathcal{M}_{\underline{\Psi}})$, with $\underline{\Psi} \in \Pi_4$, such that,*

- *The interconnection $P \star \Delta$ is well-posed for all $\Delta \in \mathbf{\Delta}$.*
- *The IQC in Definition 6 is satisfied for all $\Delta \in \mathbf{\Delta}$.*
- *The primal hard IQC $(\Psi, \mathcal{M}_\Psi)$ has a hard $(\Theta, J_{\mathbf{n}_z, \mathbf{n}_w})$ factorization.*

Suppose that **either** of the two statements hold for some $\mathcal{P} = \mathcal{P}^* \succeq 0 \in \Pi_4$.

- $$\begin{bmatrix} \mathcal{T}^* \mathcal{P} \mathcal{A} + (*) & \mathcal{T}^* \mathcal{P} \mathcal{B} \\ \mathcal{B}^* \mathcal{P} \mathcal{T} & \epsilon I \end{bmatrix} + \begin{bmatrix} \mathcal{C}^* \\ \mathcal{D}^* \end{bmatrix} \mathcal{V} \begin{bmatrix} \mathcal{C} & \mathcal{D} \end{bmatrix} \preceq 0$$
- $$\begin{bmatrix} \underline{\mathcal{T}}^* \mathcal{P} \underline{\mathcal{A}} + (*) & \underline{\mathcal{T}}^* \mathcal{P} \underline{\mathcal{B}} \\ \underline{\mathcal{B}}^* \mathcal{P} \underline{\mathcal{T}} & \epsilon I \end{bmatrix} + \begin{bmatrix} \underline{\mathcal{C}}^* \\ \underline{\mathcal{D}}^* \end{bmatrix} \underline{\mathcal{V}} \begin{bmatrix} \underline{\mathcal{C}} & \underline{\mathcal{D}} \end{bmatrix} \preceq 0 \quad (10)$$

Then for any $w, \underline{w} \in L_{e,[a,b]}$, satisfies either

$$\Psi \begin{bmatrix} P \\ I \end{bmatrix} w := \left\{ \begin{bmatrix} \mathcal{T} \dot{\mathbf{x}}(t) \\ v(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ w(t) \end{bmatrix} \right.$$

or

$$\underline{\Psi} \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w} := \left\{ \begin{bmatrix} \underline{\mathcal{T}} \dot{\mathbf{x}}(t) \\ v(t) \end{bmatrix} = \begin{bmatrix} \underline{\mathcal{A}} & \underline{\mathcal{B}} \\ \underline{\mathcal{C}} & \underline{\mathcal{D}} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ \underline{w}(t) \end{bmatrix} \right. \quad (11)$$

Then the interconnection $P \star \Delta$ with $\mathbf{x}(0) = 0$ is robustly performing.

Proof. See Appendix A.3 ■

Remark 1. *Note that we do not require information about the dual uncertainty $\underline{\Delta}$, only the primal uncertainty Δ .*

The next section applies it to the dual closed-loop realization and introduces the change of variables needed for convex controller synthesis.

6. Convex Controller Synthesis (Dual System)

Theorem 4 provides a robust-performance certificate for PIE systems in terms of the dual IQC. For analysis, this merely reformulates the primal certificate, since primal-dual equivalence preserves feasibility. The benefit of the dual formulation therefore appears in controller synthesis. Both controller constructions below use the same synthesis LPI, obtained by applying (10) to the corresponding closed-loop PIE. The choice of dual uncertainty filter changes only the realization of the controller recovered from this LPI. Related applications of IQC duality include the robust-estimation result of Scherer and Köse [33], and the disturbance-feedforward synthesis results of Köse and Scherer [19] and Venkataraman and Seiler [20].

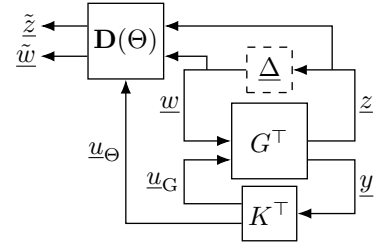


Figure 5: Dual controller interconnection.

6.1. Primal and Dual Systems

Let G , with zero initial condition $\mathbf{x}(0) = 0$, have input w , control input u , and output z . Its primal and dual PIE state-space realizations are summarized in the corresponding system-component boxes. If $\Theta \neq I$, assume that the signals entering its dynamic part are available.

6.2. State Feedback LPI

We now state the main synthesis condition. On the dual realization, the change of variables represented by $\mathcal{Z}$ makes the controller-dependent terms affine in the decision operators. Feasibility therefore returns both a robust-performance certificate and the gains of the primal controller.

Corollary 1. *Let G be the controlled PIE plant defined in Eq. (2). Let $\Delta \in \mathbf{\Delta}$ and exist a primal (hard) IQC $(\Psi, \mathcal{V})$ that has the hard $(\Theta, J_{\mathbf{n}_z, \mathbf{n}_w})$ factorization. Assume that*

$$\begin{aligned}
G^\top &:= \left\{ \begin{bmatrix} \mathcal{T}_G^* \dot{\mathbf{x}}_G(t) \\ \underline{z}(t) \\ \underline{y}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_G^* & \mathcal{C}_G^* & I \\ \mathcal{B}_G^* & \mathcal{D}_G^* & 0 \\ \mathcal{B}_u^* & \mathcal{D}_{zu}^* & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_G(t) \\ \underline{w}(t) \\ \underline{u}_G(t) \end{bmatrix} \right. & K^\top &:= \left\{ \begin{bmatrix} \underline{u}_G(t) \\ \underline{u}_\Theta(t) \end{bmatrix} = \begin{bmatrix} \mathcal{K}_G^* \\ \mathcal{K}_\Theta^* \end{bmatrix} \underline{y}(t) \right. \\
\mathbf{D}(\Theta) &:= \left\{ \begin{bmatrix} \mathcal{T}^* \dot{\mathbf{x}}_\Theta(t) \\ \underline{v}(t) \\ \underline{u}_\Theta(t) \end{bmatrix} = \begin{bmatrix} \underline{\mathcal{A}} & \underline{\mathcal{B}} & I \\ \underline{\mathcal{C}} & \underline{\mathcal{D}} & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_\Theta(t) \\ \underline{v}(t) \\ \underline{u}_\Theta(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}^* - \mathcal{C}^* \hat{\mathbf{J}} \mathbf{D}(\mathcal{D}) \hat{\mathbf{J}}^* \mathcal{B}^*, & \mathcal{C}^* \hat{\mathbf{J}} \mathbf{D}(\mathcal{D}) & I \\ -\mathbf{D}(\mathcal{D}) \hat{\mathbf{J}}^* \mathcal{B}^* & \mathbf{D}(\mathcal{D}) & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_\Theta(t) \\ \underline{v}(t) \\ \underline{u}_\Theta(t) \end{bmatrix} \right. \\
\mathbf{D}(\Theta) \begin{bmatrix} (K \star G)^\top \\ I \end{bmatrix} &:= \left\{ \begin{bmatrix} \mathcal{T}_G^* \dot{\mathbf{x}}_G(t) \\ \mathcal{T}^* \dot{\mathbf{x}}_\Theta(t) \\ \underline{\tilde{z}}(t) \\ \underline{\tilde{w}}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_G^* + \mathcal{K}_G^* \mathcal{B}_u^* & 0 & \mathcal{C}_G^* + \mathcal{K}_G^* \mathcal{D}_{zu}^* \\ \underline{\mathcal{B}}_1 \mathcal{B}_G^* + \mathcal{K}_\Theta^* \mathcal{B}_u^* & \underline{\mathcal{A}} & \underline{\mathcal{B}}_1 \mathcal{D}_G^* + \underline{\mathcal{B}}_2 + \mathcal{K}_\Theta^* \mathcal{D}_{zu}^* \\ \underline{\mathcal{D}}_{11} \mathcal{B}_G^* & \underline{\mathcal{C}}_1 & \underline{\mathcal{D}}_{11} \mathcal{D}_G^* + \underline{\mathcal{D}}_{12} \\ \underline{\mathcal{D}}_{21} \mathcal{B}_G^* & \underline{\mathcal{C}}_2 & \underline{\mathcal{D}}_{21} \mathcal{D}_G^* + \underline{\mathcal{D}}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{x}_G(t) \\ \mathbf{x}_\Theta(t) \\ \underline{w}(t) \end{bmatrix} \right. & (12)
\end{aligned}$$

Figure 4: Dual component realizations and full augmented interconnection for state feedback with dynamic filtering, expressed in terms of the primal PI operators.

- The interconnection $K \star P \star \Delta$ is well-posed for all $\Delta \in \mathbf{\Delta}$.
- The IQC in Definition 6 is satisfied for all $\Delta \in \mathbf{\Delta}$.

Moreover there exist, $\mathcal{Z}, \mathcal{P} \in \Pi_4$ with $\mathcal{P} = \mathcal{P}^* \succ 0$, such that the following statement holds

$$\begin{aligned}
& \left[\begin{array}{cc} \mathcal{T}^* \mathcal{P} \mathcal{A}_0 + \mathcal{T}^* \mathcal{Z} \begin{bmatrix} \mathcal{B}_u^* & 0 \end{bmatrix} + (*) & \mathcal{T}^* \mathcal{P} \mathcal{B}_0 + \mathcal{T}^* \mathcal{Z} \mathcal{D}_{zu}^* \\ (*) & \mathcal{I} \end{array} \right]_{\in I} \\
& + \begin{bmatrix} \underline{\mathcal{C}}^* \\ \underline{\mathcal{D}}^* \end{bmatrix} J_{\mathbf{n}_w, \mathbf{n}_z} \begin{bmatrix} \underline{\mathcal{C}} & \underline{\mathcal{D}} \end{bmatrix} \preceq 0.
\end{aligned} \quad (13)$$

where

$$\mathbf{D}(\Theta) \begin{bmatrix} 0 \star G^\top \\ I \end{bmatrix} \underline{w} := \left\{ \begin{bmatrix} \mathcal{T} \dot{\underline{x}}(t) \\ \underline{v}(t) \end{bmatrix} = \begin{bmatrix} \underline{\mathcal{A}}_0 & \underline{\mathcal{B}}_0 \\ \underline{\mathcal{C}} & \underline{\mathcal{D}} \end{bmatrix} \begin{bmatrix} \underline{x}(t) \\ \underline{w}(t) \end{bmatrix} \right. \quad (14)$$

and recover the controller gains from

$$[\mathcal{K}_G \quad \mathcal{K}_\Theta] = \mathcal{Z}^* \mathcal{P}^{-1}.$$

Assume that $\mathcal{D}_{21} \mathcal{D}_G + \mathcal{D}_{22}$ is invertible, with

$$(\mathcal{D}_{21} \mathcal{D}_G + \mathcal{D}_{22})^{-1} \in \Pi_4$$

Then the interconnection $K \star P \star \Delta$ with $\mathbf{x}(0) = 0$ is robustly stable and robust performance is achieved.

Proof. See Appendix A.4. ■

The operator $\mathcal{P}$ certifies the dual closed loop, while $\mathcal{Z}$ contains the transformed controller gains. The recovery formula maps these decision operators back to the controller acting on the primal plant and filter states. The invertibility assumption ensures that the resulting algebraic interconnection is well-posed.

6.3. $\Psi_\Delta = I$ controller realization

When $\Psi_\Delta = I$ and $\underline{\Psi}_\Delta = I$, no multiplier state is required in the controller. For a state-feedback operator $\mathcal{K}_G$,

set $u(t) = \mathcal{K}_G \mathbf{x}_G(t)$. The resulting closed-loop PIE $K \star G$ is

$$\begin{bmatrix} \partial_t(\mathcal{T}_G \mathbf{x}_G)(t) \\ z(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_G + \mathcal{B}_u \mathcal{K}_G & \mathcal{B}_G \\ \mathcal{C}_G + \mathcal{D}_{zu} \mathcal{K}_G & \mathcal{D}_G \end{bmatrix} \begin{bmatrix} \mathbf{x}_G(t) \\ w(t) \end{bmatrix}.$$

Thus the identity-filter case requires no additional controller state. The controller is memoryless in time and acts directly on the measured PIE state.

6.4. Dynamic-filter controller realization

Dynamic-IQC feedback constructions with additional controller memory have also been studied for uncertain systems with state delay [24]. Here, assume that the uncertainty Δ is known and that $z_\Delta, w_\Delta = \Delta z_\Delta$, and w_p are available to the controller. For $\underline{\Psi}_\Delta \neq I$, find the realization of $\Psi_\Delta = \mathbf{D}(\underline{\Psi}_\Delta)$,

$$\begin{bmatrix} \mathcal{T}_{\Psi_\Delta} \dot{\mathbf{x}}_{\Psi_\Delta}(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_{\Psi_\Delta} & \mathcal{B}_{\Psi_\Delta}^z & \mathcal{B}_{\Psi_\Delta}^w \\ I & 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_{\Psi_\Delta}(t) \\ z_\Delta(t) \\ w_\Delta(t) \end{bmatrix}.$$

We impose the zero initial condition $\mathbf{x}_{\Psi_\Delta}(0) = 0$. The second block row gives $y(t) = \mathbf{x}_{\Psi_\Delta}(t)$. The complete controller has the PIE realization

$$\begin{bmatrix} \mathcal{T}_{\Psi_\Delta} \dot{\mathbf{x}}_{\Psi_\Delta}(t) \\ u(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_{\Psi_\Delta} & 0 & \mathcal{B}_{\Psi_\Delta}^z & \mathcal{B}_{\Psi_\Delta}^w \\ \mathcal{K}_\Psi & \mathcal{K}_G & 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_{\Psi_\Delta}(t) \\ \mathbf{x}(t) \\ z_\Delta(t) \\ w_\Delta(t) \end{bmatrix}.$$

Hence $\mathbf{x}_{\Psi_\Delta}$ is the controller state. The plant state $\mathbf{x}$ and the signals that enter the dynamic part of the filter are external controller inputs, while u is its output. This realization makes explicit the signal-availability assumption stated in Section 4.

7. Numerical Examples

We consider PDE examples with repeated parametric and sector-bounded uncertainties.

7.1. Note on loop shifting with $\mathcal{J}$

In our examples we perform the loop shift $\frac{\partial}{\partial s} \circ \Delta$ on distributed uncertainty channels entering the dynamics. The effect of this loop shift can be seen from $\dot{\mathbf{v}} = \mathbf{v}_{ss} + \delta^2 \mathbf{v}$, with $\mathbf{v}(t, 0) = 0$ and $|\delta| \leq \alpha$. Representing δ^2 by two repeated channels and spatially differentiating their inputs and outputs gives

$$\begin{aligned}\dot{\mathbf{v}} &= \mathbf{v}_{ss} + \mathcal{J}w_{\Delta,2}, \\ z_{\Delta,1} &= \mathbf{v}_s, \quad z_{\Delta,2} = w_{\Delta,1}, \\ w_{\Delta,i} &= \delta z_{\Delta,i}, \quad i \in \{1, 2\}.\end{aligned}$$

Define the spatial integration operator $\mathcal{J} \in \Pi_3$ by $(\mathcal{J}f)(t, s) := \int_0^s f(t, \theta) d\theta$. Closing the loop recovers $\mathcal{J}w_{\Delta,2} = \delta^2 \mathcal{J}\mathbf{v}_s = \delta^2 \mathbf{v}$. The feedthrough $z_{\Delta,2} = w_{\Delta,1}$ remains unfiltered, since filtering it would instead produce the different term $\delta^2 \mathcal{J}\mathbf{v}$. This realization substantially improves feasibility in our computations, possibly because it better aligns with the fundamental PIE state. We do not have a general analytical explanation and therefore treat this benefit as empirical and realization- and uncertainty-dependent.

7.2. Primal–dual analysis with repeated real uncertainty

We begin with the finite-dimensional ODE benchmark of Veenman et al. [32, Sec. 8.1]. We copy its temporal filter basis, and performance setup, and apply the same construction to a pair of coupled heat equations by lifting the filter over the spatial domain. For the static part we can use general PI operators. This multiplier setup aligns with the multiplier class in [17, Lem. 10]. We use the same finite-dimensional matrices as in the ODE benchmark, but apply them on distributed signals, and add a diffusion diffusion to obtain the coupled PDE. Let $s \in [0, 1]$ and

$$\begin{aligned}A &= \begin{bmatrix} -2 & -3 \\ 1 & 1 \end{bmatrix} + \frac{\pi^2}{2}I, \quad B_{\Delta} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad B_p = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \\ C_{\Delta} &= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad D_{\Delta} = \begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix}, \quad D_{\Delta p} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \\ C_p &= \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D_{p\Delta} = \begin{bmatrix} 0 & 1 \end{bmatrix}.\end{aligned}$$

Consider

$$\begin{aligned}\dot{\mathbf{v}} &= \frac{1}{10}\mathbf{v}_{ss} + A\mathbf{v} + B_{\Delta}\mathcal{J}w_{\Delta} + sB_p w_p, \quad \mathbf{v}(t, 0) = 0, \quad \mathbf{v}_s(t, 1) \stackrel{\Psi_{\Delta}}{=} 0, \\ z_{\Delta} &= C_{\Delta}\mathbf{v}_s + D_{\Delta\Delta}w_{\Delta} + D_{\Delta p}w_p, \quad w_{\Delta} = \delta z_{\Delta}, \quad |\delta| \leq \alpha, \quad \mathcal{V}_{\Delta} = \begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}^* \begin{bmatrix} 0 & \underline{\mathcal{S}}_{\Delta} \\ \underline{\mathcal{S}}_{\Delta} & 0 \end{bmatrix} \begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}, \\ z_p &= 3 \int_0^1 s (C_p \mathbf{v} + D_{p\Delta} \mathcal{J}w_{\Delta}) ds.\end{aligned}$$

The same real scalar δ is repeated over both channels and all spatial locations, and $\Delta = \delta I$ is self-adjoint. We therefore use the same multiplier family in the two analyses,

$$\begin{aligned}\mathcal{V}_{\Delta}\{\alpha^2 \underline{\mathcal{Q}}_{\Delta}, \alpha \underline{\mathcal{S}}_{\Delta}, -\underline{\mathcal{Q}}_{\Delta}\}, \quad \underline{\mathcal{Q}}_{\Delta} \succeq 0, \quad \underline{\mathcal{S}}_{\Delta}^* &= -\underline{\mathcal{S}}_{\Delta}, \\ \mathcal{V}_{\Delta}\{\alpha^2 \underline{\mathcal{Q}}_{\Delta}, \alpha \underline{\mathcal{S}}_{\Delta}, -\underline{\mathcal{Q}}_{\Delta}\}, \quad \underline{\mathcal{Q}}_{\Delta} \succeq 0, \quad \underline{\mathcal{S}}_{\Delta}^* &= -\underline{\mathcal{S}}_{\Delta}.\end{aligned}$$

Table 1: One-shot computation time for each temporal filter order. Times are to be inserted from the existing timing runs for the unshifted, unit-diffusion model.

Filter order ν	Primal (s)	Dual (s)
0	—	—
1	—	—
2	—	—
3	—	—

This is the situation described in Section 5: because $\Delta = \Delta^*$, the primal and dual certificates can use the same dynamic filter and repeated-real IQC structure, with separate decision operators on the two sides.

The dynamic uncertainty filter is constructed from the stable temporal basis

$$\psi_{\nu}(p) = \begin{bmatrix} 1 & \frac{-\rho}{p-\rho} & \cdots & \left(\frac{-\rho}{p-\rho}\right)^{\nu} \end{bmatrix}^T, \quad \rho < 0,$$

where p denotes the Laplace variable. To lift this finite-dimensional filter to the spatial signal space, its state-space matrices are placed in the pointwise R_0 blocks of the corresponding PI operators, while the integral kernels R_1 and R_2 are set to zero. Thus the same temporal filter acts independently at every $s \in [0, 1]$; identical lifted copies are used on the z_{Δ} and w_{Δ} channels. The gain from w_p to z_p is tested with the primal and dual performance multipliers $\mathcal{V}_p\{1, 0, -\gamma^2\}$ and $\mathcal{V}_p\{1, 0, -\gamma^2\}$, respectively.

For each uncertainty radius and filter order, the primal and dual LPs are solved on the same pole grid. Figure 6 shows their optimized gains and pointwise difference. The computation times for the individual filter orders are reported in Table 1.

7.3. Boundary control with sector-bounded uncertainty

The following dual IQC is used for both nonlinear PDEs below.

Lemma 1 (Dual sector-bound IQC). *Suppose that $(\Delta z_{\Delta})(t, s) = \phi(z_{\Delta}(t, s))$ and*

$$\alpha z^2 \leq z\phi(z) \leq \beta z^2, \quad z \in \mathbb{R},$$

where $\alpha < \beta$. Then Δ satisfies the dual hard IQC with $\underline{\Psi}_{\Delta} = I$ and

$$\mathcal{V}_{\Delta} = \begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}^* \begin{bmatrix} 0 & \underline{\mathcal{S}}_{\Delta} \\ \underline{\mathcal{S}}_{\Delta} & 0 \end{bmatrix} \begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}, \quad (15)$$

where $\underline{\mathcal{S}}_{\Delta} \in \Pi_3$ is the positive multiplication operator

$$(\underline{\mathcal{S}}_{\Delta} \mathbf{x})(t, s) = \underline{\mathcal{S}}_0(s) \mathbf{x}(t, s), \quad \underline{\mathcal{S}}_0(s) > 0. \quad (16)$$

Proof. See Appendix A.5. ■

Optimal and robust boundary control of the sine–Gordon equation has also been studied on finite horizons [34].

Consider boundary control of the reaction–diffusion (RD) and damped wave (W) equations

$$\begin{aligned}\dot{\mathbf{v}}_{\text{RD}} &= \mathbf{v}_{\text{RD},ss} + \lambda_{\text{RD}} \sin(\mathbf{v}_{\text{RD}}) + sw_{\text{p}}, \\ \ddot{\mathbf{v}}_{\text{W}} &= \mathbf{v}_{\text{W},ss} - 2\dot{\mathbf{v}}_{\text{W}} + \lambda_{\text{W}} \sin(\mathbf{v}_{\text{W}}) + sw_{\text{p}},\end{aligned}$$

with, for $i \in \{\text{RD}, \text{W}\}$,

$$\mathbf{v}_i(t, 0) = 0, \quad \mathbf{v}_{i,s}(t, 1) = x_i(t), \quad \dot{x}_i(t) = u_i(t), \quad z_{\text{p}} = \int_0^1 \mathbf{v}_i(t, s) ds.$$

For synthesis, set

$$z_{\Delta,i} = \mathbf{v}_{i,s}, \quad w_{\Delta,i} = \cos(\mathcal{J}z_{\Delta,i})z_{\Delta,i},$$

and replace $\lambda_i \sin(\mathbf{v}_i)$ by $\lambda_i \mathcal{J}w_{\Delta,i}$. Since $\mathcal{J}z_{\Delta,i} = \mathbf{v}_i$, this is an equivalent realization; explicitly,

$$\mathcal{J}w_{\Delta,i} = \mathcal{J}(\cos(\mathbf{v}_i)\mathbf{v}_{i,s}) = \sin(\mathbf{v}_i) - \sin(\mathbf{v}_i(t, 0)) = \sin(\mathbf{v}_i).$$

Also, $sw_{\text{p}}(t) = (\mathcal{J}w_{\text{p}})(t, s)$. Moreover,

$$-z_{\Delta,i}^2 \leq z_{\Delta,i}w_{\Delta,i} \leq z_{\Delta,i}^2,$$

so the same $[-1, 1]$ sector IQC applies to both plants. We use Lemma 1 with $\alpha = -1$, $\beta = 1$, the performance multiplier $\mathcal{Y}_{\text{p}}\{1, 0, -\gamma^2 I_2\}$, and $\|\mathcal{K}_i\| \leq 10^4$. As discussed in Section 5, the primal and dual analyses use the same sector-IQC structure, while the synthesis condition uses its underlined dual counterpart.

The computations use $\lambda_{\text{RD}} = 5$ and $\lambda_{\text{W}} = 3$. The state simulations use $w_{\text{p}}(t) = 8 \cos(t) \mathbf{1}_{[\pi, 4\pi]}(t)$. For the separately reported wave certificate, the performance output is specialized to $z_{\text{p}}(t) = \int_0^1 \mathbf{v}_{\text{W}}(t, s) ds$.

For this scalar wave objective, solving (13) yields the certified bound $\gamma = 0.45$ for the closed-loop interconnection $P_K \star \Delta$. The corresponding state-feedback controller is

$$u(t) = -15.46x(t) + \int_0^1 (2Q_1(s)\mathbf{v}_{\text{W},ss}(t, s) + Q_2(s)\dot{\mathbf{v}}_{\text{W}}(t, s)) ds$$

where

$$\begin{aligned}Q_1(s) &= 200.61s^{12} - 1537.17s^{11} + 5033.27s^{10} - 9096.06s^9 \\ &\quad + 9933.67s^8 - 6788.31s^7 + 2916.66s^6 - 700.58s^5 \\ &\quad - 82.97s^4 + 161.05s^3 - 40.98s^2 + 3.72s - 7.6113, \\ Q_2(s) &= 203.29s^{12} - 1309.04s^{11} + 3721.08s^{10} - 5890.06s^9 \\ &\quad + 5519.18s^8 - 3072.06s^7 + 942.76s^6 + 1.51s^5 \\ &\quad - 274.86s^4 + 166.68s^3 - 21.49s^2 + 11.76s - 0.07.\end{aligned}$$

For these parameter values, the dual synthesis LPI produces stabilizing boundary controllers. Figure 7 shows the resulting open- and closed-loop responses.

8. Connection to finite-dimensional dynamic multipliers

This section connects the duality framework developed above with the established theory of finite-dimensional dynamic IQC multipliers. For rational multipliers satisfying the strict positive–negative (PN) conditions, Seiler [15] showed that a J -spectral factorization provides a stable and stably invertible hard factorization, thereby converting the associated soft, infinite-horizon IQC into a time-domain constraint that holds on every finite horizon. As discussed in the linear parameter-varying (LPV) setting by Venkataraman and Seiler [20], this J -spectral factorization satisfies the filter assumptions needed to apply the duality theory. The first part of this section reviews this connection between strict PN multipliers, hard factorizations, and duality. The second part illustrates the construction using a delayed firing-rate model.

8.1. Positive–Negative Multipliers

Using the terminology of Definition 8, we denote the J -spectral factor by Θ . The following result restates [15, Thm. 4] in this notation. Moreover, for finite dimensional multipliers we can define $\Pi = \Psi^* V \Psi \in \mathbb{R}L_{\infty}$, where $\Psi(j\omega)^* = \Psi(-j\omega)^{\top}$.

Theorem 5. *Let $\Pi = \Pi^* \in \mathbb{R}L_{\infty}^{(n_z+n_w) \times (n_z+n_w)}$ where $\Pi_{11} \in \mathbb{R}L_{\infty}^{n_z \times n_z}$ and $\Pi_{22} \in \mathbb{R}L_{\infty}^{n_w \times n_w}$. If $\Pi_{11}(j\omega) \succ 0, \Pi_{22}(j\omega) \prec 0$, for all $\omega \in \mathbb{R} \cup \{\infty\}$, then*

1. Π admits a J_{n_z, n_w} -spectral factorization $\Pi = \Theta^* J_{n_z, n_w} \Theta$, where $\Theta, \Theta^{-1} \in \mathbb{R}H_{\infty}^{(n_z+n_w) \times (n_z+n_w)}$.
2. (Θ, J_{n_z, n_w}) is a hard factorization of Π .

Pfifer and Seiler [35, App. B] give a Riccati-based construction of J -spectral factors for strict PN multipliers, using the rational J -spectral factorization results of Meinsma [36].

The distinction between a frequency-domain factorization identity and the finite-horizon conditions used in Definition 8 is relevant here. Preservation of both graph and inverse-graph inequalities is studied through doubly-hard factorizations [25]. Finite-horizon IQCs with terminal costs provide another connection to dissipativity [26], while multiplier storage can reduce conservatism [27, 28]. These results provide context for the finite-horizon assumptions imposed in Definition 8. They do not by themselves establish the truncated quadratic-form identity in that definition.

8.2. Example. Generation of β -Oscillations in the Subthalamic Nucleus-Globus Pallidus Network

Consider the delayed firing-rate model of Nevado Holgado, Terry, and Bogacz [37, Eq. (5)], and define the stim-

ulation input $\tilde{u}(t) := I_e \text{Stim}(t)$.

$$\begin{aligned}\tau_S \dot{\text{STN}}(t) &= F_S(-w_{GS} \text{GP}(t - \tau_{GS}) \\ &\quad + w_{CS} \text{Ctx}) - \text{STN}(t) + \tilde{u}(t), \\ \tau_G \dot{\text{GP}}(t) &= F_G(w_{SG} \text{STN}(t - \tau_{SG}) \\ &\quad - w_{GG} \text{GP}(t - \tau_{GG}) \\ &\quad - w_{XG} \text{Str}) - \text{GP}(t).\end{aligned}\quad (17)$$

Let STN and GP denote the physical firing rates of the SubThalamic Nucleus (STN) and Globus Pallidus pars Externa (GPe). The activation functions are

$$F_i(q) = \frac{M_i}{1 + \exp(-4q/M_i)(M_i - B_i)/B_i)}, \quad i \in \{S, G\}, \quad (18)$$

so $F_i(0) = B_i$. Define the sigmoid midpoint and its centered form as

$$\begin{aligned}c_i &:= \frac{M_i}{4} \ln\left(\frac{M_i - B_i}{B_i}\right), \\ \sigma_i(q) &:= \frac{M_i}{1 + \exp(-4q/M_i)} - \frac{M_i}{2} = \frac{M_i}{2} \tanh\left(\frac{2q}{M_i}\right).\end{aligned}\quad (19)$$

Then $F_i(q) = \sigma_i(q - c_i) + M_i/2$. Define the centered states

$$\tilde{x}_S := \text{STN} - \frac{M_S}{2}, \quad \tilde{x}_G := \text{GP} - \frac{M_G}{2},$$

and offsets

$$\begin{aligned}b_S &:= w_{CS} \text{Ctx} - \frac{w_{GS} M_G}{2} - c_S, \\ b_G &:= \frac{w_{SG} M_S}{2} - \frac{w_{GG} M_G}{2} - w_{XG} \text{Str} - c_G.\end{aligned}$$

This yields a model in which the nonlinearities σ_S and σ_G are centered and satisfy $\sigma_i(0) = 0$,

$$\begin{aligned}\tilde{z}_S(t) &:= -w_{GS} \tilde{x}_G(t - \tau_{GS}) + b_S, \\ \tilde{z}_G(t) &:= w_{SG} \tilde{x}_S(t - \tau_{SG}) - w_{GG} \tilde{x}_G(t - \tau_{GG}) + b_G, \\ \tau_S \dot{\tilde{x}}_S(t) &= \sigma_S(\tilde{z}_S(t)) - \tilde{x}_S(t) + \tilde{u}(t), \\ \tau_G \dot{\tilde{x}}_G(t) &= \sigma_G(\tilde{z}_G(t)) - \tilde{x}_G(t).\end{aligned}\quad (20)$$

Here τ_{ij} and w_{ij} denote the transmission delay and synaptic weight, respectively, from population i to population j . τ_S and τ_G denote the time constants for the STN and GPe populations, respectively. To formulate the control objective around a desired operating point, we introduce the equilibrium $(\tilde{x}_S^*, \tilde{x}_G^*)$ satisfying

$$\begin{aligned}\tilde{z}_S^* &:= -w_{GS} \tilde{x}_G^* + b_S, \\ \tilde{z}_G^* &:= w_{SG} \tilde{x}_S^* - w_{GG} \tilde{x}_G^* + b_G, \\ \tilde{x}_S^* &= \sigma_S(\tilde{z}_S^*) + \tilde{u}^*, \\ \tilde{x}_G^* &= \sigma_G(\tilde{z}_G^*).\end{aligned}$$

For $i \in \{S, G\}$, introduce the deviation variables

$$x_i(t) := \tilde{x}_i(t) - \tilde{x}_i^*, \quad z_i(t) := \tilde{z}_i(t) - \tilde{z}_i^*, \quad u(t) := \tilde{u}(t) - \tilde{u}^*,$$

and define the shifted nonlinearities

$$\delta_i(v) := \sigma_i(\tilde{z}_i^* + v) - \sigma_i(\tilde{z}_i^*).$$

The resulting system is a nonlinear DDE with the equilibrium shifted to the origin. Following the delay representation of Peet [38], we introduce the delayed states $\phi_{ij}(t, s) := x_i(t - \tau_{ij}s)$ for $(i, j) \in \mathcal{I}_D$ and $s \in [0, 1]$, where $\mathcal{I}_D := \{(G, S), (S, G), (G, G)\}$. This yields the equivalent ODE-PDE representation,

$$\begin{aligned}z_S(t) &:= -w_{GS} \phi_{GS}(t, 1), \\ z_G(t) &:= w_{SG} \phi_{SG}(t, 1) - w_{GG} \phi_{GG}(t, 1), \\ \tau_S \dot{x}_S(t) &= \delta_S(z_S(t)) - x_S(t) + u(t), \\ \tau_G \dot{x}_G(t) &= \delta_G(z_G(t)) - x_G(t), \\ \dot{\phi}_{ij}(t, s) &= -\frac{1}{\tau_{ij}} \phi_{ij,s}(t, s), \\ \phi_{ij}(t, 0) &= x_i(t).\end{aligned}\quad (21)$$

Thus, the equilibrium shift only affects the nonlinear operators, while leaving the linear delayed dynamics unchanged. We now turn to robustness and performance analysis.

8.3. Robust Stability Analysis

We first seek the largest sector or slope bound β for which a stimulation controller can certify robust stability of (21). Thus each feasibility test includes controller synthesis. The controller is a design variable throughout the search in β . At this stage the objective is robust stabilization, with the performance channels omitted. Write

$$z_\Delta := \begin{bmatrix} z_S \\ z_G \end{bmatrix}, \quad w_\Delta := \begin{bmatrix} \delta_S(z_S) \\ \delta_G(z_G) \end{bmatrix}.$$

The shifted nonlinearities are memoryless, time invariant, monotone, and vanish at the origin. We use a common upper bound $\beta > 0$ for both channels, first retaining only the sector information

$$0 \leq v \delta_i(v) \leq \beta v^2, \quad i \in \{S, G\}, \quad (22)$$

and then also using the slope restriction

$$0 \leq \frac{\delta_i(v_1) - \delta_i(v_2)}{v_1 - v_2} \leq \beta, \quad v_1 \neq v_2. \quad (23)$$

The slope restriction implies the sector bound by taking $v_2 = 0$. We initially test these uncertainty classes on the whole real line. Below we identify the trajectories of the actual sigmoids to which a given β applies. Here β denotes a sector or slope bound, independently of the beta frequency band in the example title.

8.3.1. Sector uncertainty

For $D := \text{diag}(d_S, d_G) \succ 0$, the sector IQC acting on $\begin{bmatrix} z_\Delta^\top & w_\Delta^\top \end{bmatrix}^\top$ has multiplier

$$\Pi_{\text{sec}} := \begin{bmatrix} 0 & \beta D \\ \beta D & -2D \end{bmatrix}. \quad (24)$$

Its quadratic form is nonnegative pointwise by (22). Applying Lemma 1 channelwise with $\alpha = 0$ directly gives the hard dual IQC

$$\underline{\Psi}_\Delta = I_4, \quad \underline{\mathcal{V}}_\Delta = \begin{bmatrix} 0 & \beta D \\ \beta D & -2D \end{bmatrix}.$$

Thus the sector synthesis uses this hard dual IQC directly, without an ϵ completion or the ARE-based factorization. For each trial β and an admissible choice of D , we seek a controller and storage operator satisfying the stability-only synthesis LPI with this IQC. Searching in β gives the largest sector bound certified by this design procedure, denoted β_{sec}^* , together with a stabilizing controller at each feasible bound. For the present parameterization, the largest certified sector bound is $\beta_{\text{sec}}^* = 0.492$.

8.3.2. Zames–Falb multiplier

Zames and Falb [39] introduced multipliers that exploit monotonicity and slope restrictions on nonlinearities. To exploit (23), we replace the static weight D by a diagonal Zames–Falb multiplier using the construction of Turner [40],

$$\begin{aligned} M(\zeta) &:= I - \text{diag}(H_S(\zeta), H_G(\zeta)), \\ H_i(\zeta) &:= \sum_{k=1}^3 \frac{\kappa_{ik}}{\zeta + a_{ik}}, \\ a_{ik} &> 0, \quad \kappa_{ik} \geq 0, \quad \sum_{k=1}^3 \frac{\kappa_{ik}}{a_{ik}} \leq 1. \end{aligned} \quad (25)$$

Each term has a nonnegative exponential impulse response, with integral weight κ_{ik}/a_{ik} . Thus $\|H_i\|_{L_1} \leq 1$, giving a scaled ZF multiplier valid for monotone, slope-restricted nonlinearities without assuming oddness [40, Thm. 4]. This matters because shifting the sigmoid to an arbitrary equilibrium generally removes its odd symmetry.

We choose the three pole rates from the population time constants and the recurrent GPe delay,

$$\begin{bmatrix} a_{i1} & a_{i2} & a_{i3} \end{bmatrix} = c_i \begin{bmatrix} \tau_G^{-1} & \tau_S^{-1} & \tau_{GG}^{-1} \end{bmatrix}, \quad c_i > 0. \quad (26)$$

The poles $-a_{ik}$ supply memory on the GPe relaxation, STN relaxation, and GPe recurrent-delay scales. The reciprocal delay is a characteristic rate, not a pole of the delay operator. The dimensionless factors c_i adjust each multiplier’s time scale relative to these physical scales. We initially take $c_S = c_G = 1$.

To relate the residues to the synaptic coupling strengths, define

$$\begin{aligned} \begin{bmatrix} h_1 & h_2 & h_3 \end{bmatrix} &:= \frac{\rho \begin{bmatrix} |w_{GS}| & |w_{SG}| & |w_{GG}| \end{bmatrix}}{|w_{GS}| + |w_{SG}| + |w_{GG}|}, \\ \kappa_{ik} &:= a_{ik} h_k, \quad 0 < \rho < 1. \end{aligned}$$

This assigns the GPe relaxation term a weight proportional to the GPe-to-STN coupling, the STN relaxation term a

weight proportional to the STN-to-GPe coupling, and the recurrent-delay term a weight proportional to the GPe self-coupling. The weights are nonnegative, and normalization preserves the ZF norm condition as the synaptic parameters change,

$$\sum_{k=1}^3 \frac{\kappa_{ik}}{a_{ik}} = \sum_{k=1}^3 h_k = \rho < 1, \quad \|H_i\|_{L_1} = \rho.$$

We take $\rho = 13/15$ to retain the prescribed kernel norm. This parameterization is a physically motivated choice of multiplier shape. It does not assert that the synaptic strengths determine the optimal IQC weights or guarantee an improved certificate. The factors c_i and ρ can be compared through an outer synthesis search. For fixed system parameters and these choices, the poles and residues remain fixed during the search in β . The resulting slope IQC is

$$\Pi_{\text{ZF}}(j\omega) := \begin{bmatrix} 0 & \beta M(j\omega)^* \\ \beta M(j\omega) & -(M(j\omega) + M(j\omega)^*) \end{bmatrix}. \quad (27)$$

With all $\kappa_{ik} = 0$, this reduces to the sector multiplier. We repeat the search in β , synthesizing a controller for each trial bound using the ZF family, and denote the largest certified bound by β_{ZF}^* . Retaining the sector solution as a candidate gives $\beta_{\text{ZF}}^* \geq \beta_{\text{sec}}^*$ when the controller and storage parameterizations include those of the sector design. These bounds describe what the chosen synthesis procedure certifies. Failure of a synthesis LPI does not establish that robust stabilization is impossible. For the present parameterization, the largest certified Zames–Falb bound is $\beta_{\text{ZF}}^* = 0.562$.

For the ZF multiplier, we use a J -spectral factorization to obtain the required hard IQC and dual filter. Since the sector bound gives $\|w_\Delta\| \leq \beta \|z_\Delta\|$, we use the PN completion as discussed in [15]

$$\Pi_\Delta := \Pi_{\text{ZF}} + \epsilon \begin{bmatrix} \beta^2 I_2 & 0 \\ 0 & -I_2 \end{bmatrix}, \quad \epsilon > 0, \quad (28)$$

The added term is a valid norm IQC. Since $M(j\omega) + M(j\omega)^* \succeq 0$, Π_Δ is strict PN and Theorem 5 supplies

$$\Pi_\Delta = \Theta_\Delta^\sim J_{2,2} \Theta_\Delta, \quad \Theta_\Delta, \Theta_\Delta^{-1} \in \mathbb{R}H_\infty.$$

For each trial β in the ZF search, select admissible multiplier weights and compute this factorization. With Θ_Δ fixed, the dual synthesis LPI in Corollary 1, with performance channels omitted, searches for a controller and storage operator certifying closed-loop robust stability. Multiplier choices can be compared outside this LPI. Joint optimization of the factorization and controller is not assumed to be convex. The resulting feasible values of β therefore belong to controller–certificate pairs.

8.3.3. Admissible trajectories

For a controller certified at a given β , the admissible sigmoid inputs follow from (19),

$$\delta'_i(v) = \text{sech}^2(2(\tilde{z}_i^* + v)/M_i).$$

Consequently the sector and slope bounds hold globally for $\beta \geq 1$. For $0 < \beta < 1$, inversion of this derivative gives

$$\delta'_i(v) \leq \beta \iff |\tilde{z}_i^* + v| \geq \frac{M_i}{2} \operatorname{arcosh}(\beta^{-1/2}).$$

To obtain an interval containing the equilibrium, remain on its side of the sigmoid midpoint. A simple sufficient trajectory bound is

$$|z_i(t)| < |\tilde{z}_i^*| - \frac{M_i}{2} \operatorname{arcosh}(\beta^{-1/2}), \quad (29)$$

for $i \in \{S, G\}$ and every $t \geq 0$. The right-hand side is positive when $\beta > \delta'_i(0)$. Throughout this interval the derivative is bounded by β , so both (22) and (23) hold. For an equilibrium below the midpoint, the less restrictive one-sided condition is

$$z_i(t) \leq -\tilde{z}_i^* - \frac{M_i}{2} \operatorname{arcosh}(\beta^{-1/2}).$$

Above the midpoint, the corresponding condition is

$$z_i(t) \geq -\tilde{z}_i^* + \frac{M_i}{2} \operatorname{arcosh}(\beta^{-1/2}).$$

These bounds constrain the delayed firing-rate combinations in (21). A larger feasible β permits a wider interval around the same equilibrium. The local IQC certificate applies via a globally slope-restricted extension agreeing with the sigmoid on this interval, and transfers to the original system for trajectories that remain there. Invariance requires a separate check.

8.4. Robust Performance Analysis

Having established the robustness limit through controller synthesis, fix a feasible $\beta < \beta_{ZF}^*$ and synthesize controllers with performance objectives at this smaller bound. This leaves margin relative to the robust-stability limit. For a local sigmoid certificate, also require $\beta > \max_i \delta'_i(0)$ so that (29) contains a neighborhood of the equilibrium. This choice is possible only if the two requirements leave a nonempty interval. Keep β fixed throughout the performance comparison so that all controllers are assessed on the same admissible trajectory range.

Let w_p be an additive disturbance in the stimulation channel. Using the same uncertainty channels and transport equations as above, the generalized plant is specified by

$$\begin{aligned} \tau_S \dot{x}_S &= w_{\Delta,1} - x_S + u + w_p, \\ \tau_G \dot{x}_G &= w_{\Delta,2} - x_G, \\ z_p &:= \begin{bmatrix} x_S \\ x_G \\ \mu u \end{bmatrix}, \quad \mu > 0. \end{aligned} \quad (30)$$

Thus the performance objective measures rejection of perturbations entering through the stimulation input. The effort penalty concerns the stimulation deviation $u = \tilde{u} - \tilde{u}^*$.

We assess the induced L_2 gain to each signal separately, using γ_S , γ_G , and γ_u for x_S , x_G , and u , respectively. To generate controllers with different effort penalties, for zero initial deviations and controller state minimize the weighted synthesis bound γ in

$$\|x_S\|_{L_2}^2 + \|x_G\|_{L_2}^2 + \mu^2 \|u\|_{L_2}^2 \leq \gamma^2 \|w_p\|_{L_2}^2. \quad (31)$$

This directly penalizes firing-rate deviations and control effort, with the performance IQC

$$\Psi_p = I_4, \quad \mathcal{V}_p = \operatorname{diag}(I_3, -\gamma^2). \quad (32)$$

For $\gamma > 0$, its $J_{3,1}$ -spectral factor $\operatorname{diag}(I_3, \gamma)$ is stably invertible and can be combined with Θ_Δ in the dual synthesis condition.

For each μ , fix an admissible uncertainty multiplier at the chosen β , compute its factorization, and use Corollary 1 to synthesize the stimulation controller while minimizing γ . Keeping the multiplier fixed in each synthesis problem avoids a joint search over its factorization and the controller. Increasing μ gives more weight to reducing stimulation effort. Decreasing it gives more weight to suppressing x_S and x_G . We will vary μ to obtain a family of controllers and then reanalyze each fixed controller with three separate performance outputs to certify

$$\begin{aligned} \|x_S\|_{L_2} &\leq \gamma_S \|w_p\|_{L_2}, \\ \|x_G\|_{L_2} &\leq \gamma_G \|w_p\|_{L_2}, \\ \|u\|_{L_2} &\leq \gamma_u \|w_p\|_{L_2}. \end{aligned}$$

Each gain bound is minimized in its own analysis problem, for the same controller, disturbance channel, and uncertainty bound β . We will report all three bounds and compare γ_u with γ_S and γ_G separately, showing how the effort–attenuation tradeoff differs between the two populations. The weighted γ serves only as the synthesis objective. The reported gains refer to the individual, unweighted signals. For the original sigmoid with $\beta < 1$, each comparison is restricted to trajectories satisfying (29). The L_2 -gain bounds alone do not establish this pointwise condition.

9. Conclusion

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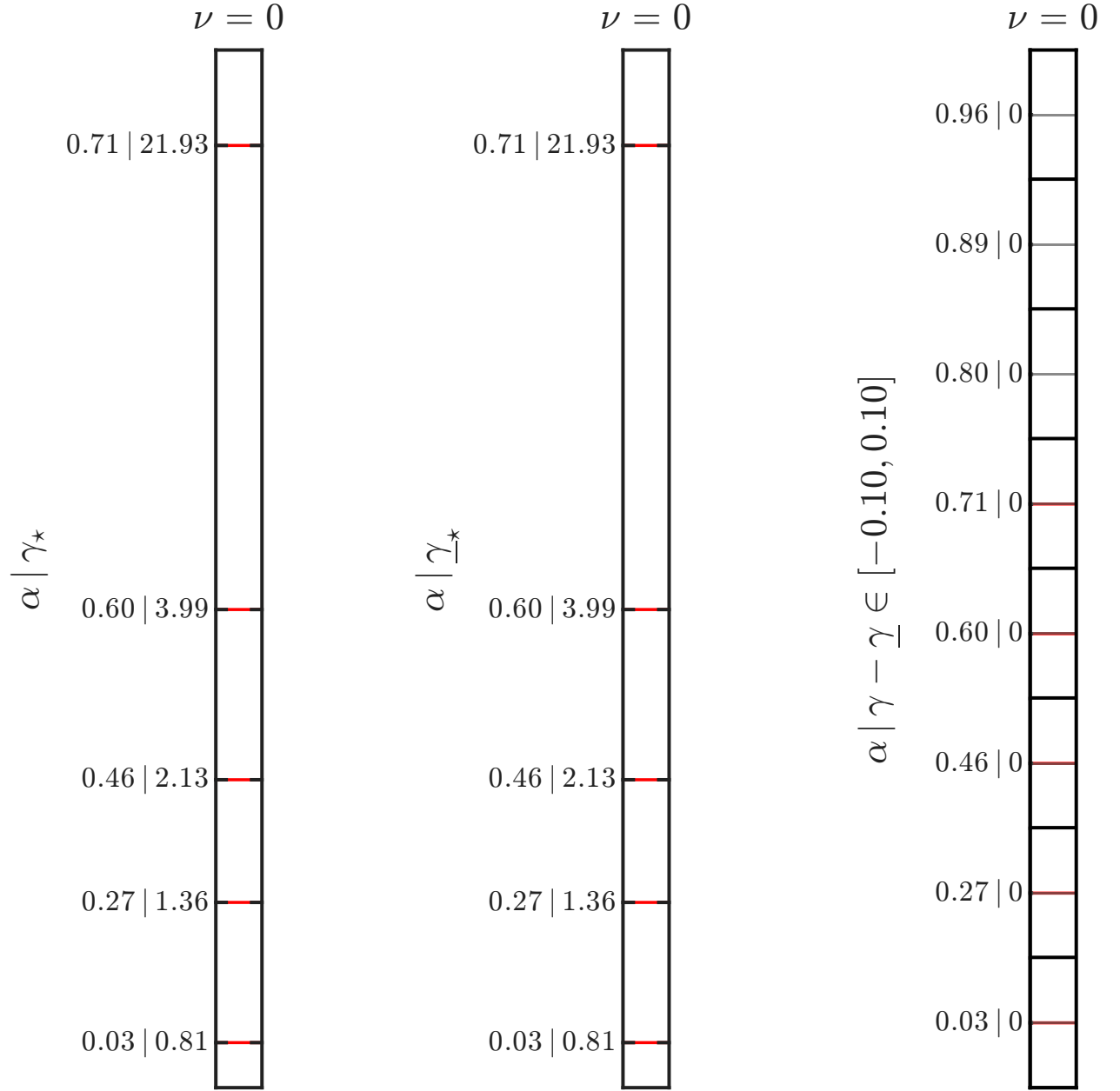


Figure 6: Reproduction of the Figure 7 sweep for the coupled heat equations: primal gain bounds (left), dual gain bounds (center), and the pointwise difference $\gamma_{\text{primal}} - \gamma_{\text{dual}}$ (right). The difference columns use the same horizontal filter-order layout as the other panels. Within each column, the uncertainty radii are stacked in equal-height strips with a common symmetric y-range and one shared scientific-notation exponent.

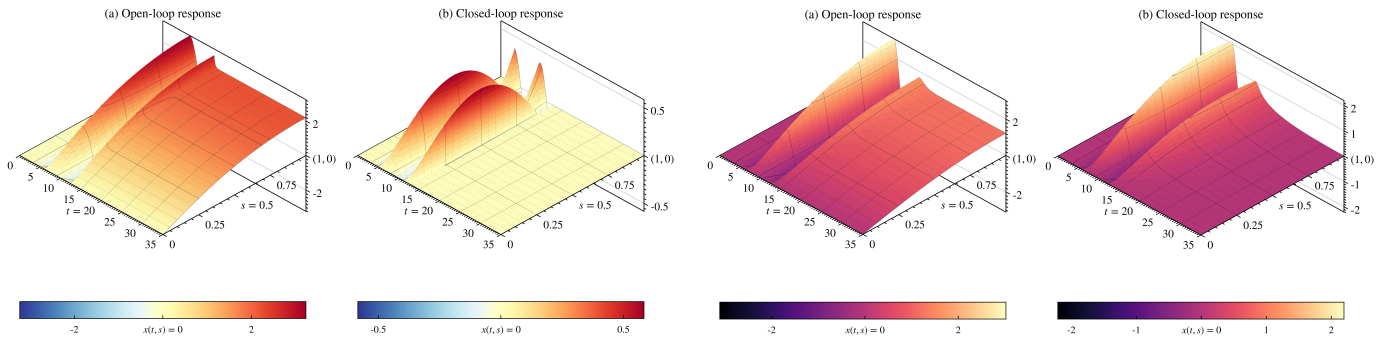


Figure 7: Open- and closed-loop state responses for the nonlinear reaction–diffusion equation (left pair) and the damped wave equation (right pair).

Appendix A. Proofs

Appendix A.1. Proof of Theorem 1

Suppose that (3) holds. This implies,

$$\left\langle p_T \Psi_\Delta \begin{bmatrix} P_\Delta \\ I \end{bmatrix} w_\Delta, p_T \mathcal{M}_{\mathcal{V}_\Delta} \Psi_\Delta \begin{bmatrix} P_\Delta \\ I \end{bmatrix} w_\Delta \right\rangle_{L[a,b]} + \left\langle p_T \Psi_p \begin{bmatrix} P_{p\Delta} \\ 0 \end{bmatrix} w_\Delta, p_T \mathcal{M}_{\mathcal{V}_p} \Psi_p \begin{bmatrix} P_{p\Delta} \\ 0 \end{bmatrix} w_\Delta \right\rangle_{L[a,b]} \leq -\epsilon \|p_T w_\Delta\|_{L[a,b]}^2.$$

By Definition 7 we can then infer,

$$\left\langle p_T \Psi_\Delta \begin{bmatrix} P_\Delta w_\Delta \\ w_\Delta \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_\Delta} \Psi_\Delta \begin{bmatrix} P_\Delta w_\Delta \\ w_\Delta \end{bmatrix} \right\rangle_{L[a,b]} \leq -\epsilon \|p_T w_\Delta\|_{L[a,b]}^2.$$

Then, using Theorem 6 in [17], we find that **robust stability** is guaranteed. Moreover, rewriting (3) as,

$$\left\langle p_T \Psi_\Delta \begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_\Delta} \Psi_\Delta \begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix} \right\rangle_{L[a,b]} + \left\langle p_T \Psi_p \begin{bmatrix} z_p \\ w_p \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_p} \Psi_p \begin{bmatrix} z_p \\ w_p \end{bmatrix} \right\rangle_{L[a,b]} \leq -\epsilon (\|p_T w_\Delta\|_{L[a,b]}^2 + \|p_T w_p\|_{L[a,b]}^2),$$

and given that Ψ_Δ and $\mathcal{M}_{\mathcal{V}_\Delta}$ satisfy the IQC in Definition 6 leads to,

$$\left\langle p_T \Psi_p \begin{bmatrix} z_p \\ w_p \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_p} \Psi_p \begin{bmatrix} z_p \\ w_p \end{bmatrix} \right\rangle_{L[a,b]} \leq -\epsilon \|p_T w_p\|_{L[a,b]}^2.$$

Thus by Definition 7 **robust performance** is achieved.

Appendix A.2. Proof of Theorem 3

1) $\implies$ 2) Assume there exists a $\rho < 1$ such that $\|p_T G_0 \tilde{w}\|_{L[a,b]} \leq \rho \|p_T \tilde{w}\|_{L[a,b]}$ for all $\tilde{w} \in L[a,b]$ and $T \geq 0$. We need to show that $\mathcal{I} \leq -\epsilon \|p_T w\|_{L[a,b]}^2$ for some $\epsilon > 0$. First since Γ is causally boundedly invertible, $\Theta \begin{bmatrix} P \\ I \end{bmatrix} w = \begin{bmatrix} G_0 \\ I \end{bmatrix} \Gamma w$, and hence $\mathcal{I} = \|p_T G_0 \Gamma w\|_{L[a,b]}^2 - \|p_T \Gamma w\|_{L[a,b]}^2$. By statement 1 and causality of G_0 ,

$$\mathcal{I} \leq -(1 - \rho^2) \|p_T \Gamma w\|_{L[a,b]}^2 \leq -\frac{1 - \rho^2}{\|\Gamma^{-1}\|^2} \|p_T w\|_{L[a,b]}^2,$$

where the second inequality follows from $\|p_T w\|_{L[a,b]} = \|p_T \Gamma^{-1} p_T \Gamma w\|_{L[a,b]} \leq \|\Gamma^{-1}\| \|p_T \Gamma w\|_{L[a,b]}$. Thus, $\epsilon = \frac{(1 - \rho^2)}{\|\Gamma^{-1}\|^2} > 0$.

2) $\implies$ 1) Assume that there exists a $\epsilon > 0$ such that $\mathcal{I} \leq -\epsilon \|p_T w\|_{L[a,b]}^2$ for every w , and every $T \geq 0$. We need to establish there exists a $\rho < 1$ such that $\|p_T G_0 \tilde{w}\|_{L[a,b]} \leq \rho \|p_T \tilde{w}\|_{L[a,b]}$ for all $\tilde{w} \in L[a,b]$ and $T \geq 0$. For arbitrary $\tilde{w}$, set $w = \Gamma^{-1} \tilde{w}$. Then

$$\|p_T G_0 \tilde{w}\|_{L[a,b]}^2 - \|p_T \tilde{w}\|_{L[a,b]}^2 \leq -\epsilon \|p_T \Gamma^{-1} \tilde{w}\|_{L[a,b]}^2.$$

Since Γ is bounded and causal, $\|p_T \tilde{w}\|_{L[a,b]} \leq \|\Gamma\| \|p_T \Gamma^{-1} \tilde{w}\|_{L[a,b]}$. Therefore,

$$\|p_T G_0 \tilde{w}\|_{L[a,b]}^2 \leq \left(1 - \frac{\epsilon}{\|\Gamma\|^2}\right) \|p_T \tilde{w}\|_{L[a,b]}^2.$$

Choose $\rho < 1$ such that $1 - \epsilon/\|\Gamma\|^2 \leq \rho^2 < 1$, then $\|p_T G_0 \tilde{w}\|_{L[a,b]} \leq \rho \|p_T \tilde{w}\|_{L[a,b]}$ for every $\tilde{w}$ and $T \geq 0$.

1) $\iff$ 3) We show that there exists a causally boundedly invertible $\underline{\Gamma}$ such that

$$\mathbf{D}(\Theta) \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w} = \begin{bmatrix} G_0^\top \\ I \end{bmatrix} \underline{\Gamma} w.$$

then we can apply the previous arguments to show 1) $\iff$ 3). Define

$$S := \Theta_{11} - \Theta_{12} \Theta_{22}^{-1} \Theta_{21}.$$

Since Θ and Θ_{22} are boundedly invertible, so is S by Schur complement. Block inversion of $\mathbf{D}(\Theta)$ gives

$$\mathbf{D}(\Theta)_{21} = (S^\top)^{-1} \Theta_{21}^\top (\Theta_{22}^\top)^{-1}, \quad \mathbf{D}(\Theta)_{22} = (S^\top)^{-1}.$$

Therefore, define

$$\underline{\Gamma} := \mathbf{D}(\Theta)_{21}P^\top + \mathbf{D}(\Theta)_{22} = (S^\top)^{-1}(I + P\Theta_{22}^{-1}\Theta_{21})^\top.$$

Now $\Gamma = \Theta_{22}(I + \Theta_{22}^{-1}\Theta_{21}P)$. Thus, invertibility of Γ and the push-through rule imply that $I + P\Theta_{22}^{-1}\Theta_{21}$, and hence $\underline{\Gamma}$, is causally boundedly invertible. Normalizing the upper block gives

$$\mathbf{D}(\Theta) \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w} = \begin{bmatrix} \underline{G}_0 \\ I \end{bmatrix} \underline{\Gamma}w, \quad \underline{G}_0 := (\Theta_{22}^\top)^{-1}(P^\top \underline{\Gamma}^{-1} + \Theta_{12}^\top).$$

Using

$$\underline{\Gamma}^{-\top} = S(I + P\Theta_{22}^{-1}\Theta_{21})^{-1}, \quad (I + P\Theta_{22}^{-1}\Theta_{21})^{-1}P\Theta_{22}^{-1} = P\Gamma^{-1},$$

where the second identity is the push-through rule, yields

$$\begin{aligned} \underline{G}_0 &= (\Theta_{22}^\top)^{-1}(P^\top \underline{\Gamma}^{-1} + \Theta_{12}^\top) \\ &= \left(\underline{\Gamma}^{-\top} P\Theta_{22}^{-1} + \Theta_{12}\Theta_{22}^{-1} \right)^\top \\ &= [S(I + P\Theta_{22}^{-1}\Theta_{21})^{-1}P\Theta_{22}^{-1} + \Theta_{12}\Theta_{22}^{-1}]^\top \\ &= [SP\Gamma^{-1} + \Theta_{12}\Theta_{22}^{-1}]^\top \\ &= [(SP + \Theta_{12}\Theta_{22}^{-1}\Gamma)\Gamma^{-1}]^\top. \end{aligned}$$

We now show that the first part reduces to

$$\begin{aligned} SP + \Theta_{12}\Theta_{22}^{-1}\Gamma &= \Theta_{11}P - \Theta_{12}\Theta_{22}^{-1}\Theta_{21}P \\ &\quad + \Theta_{12}\Theta_{22}^{-1}\Theta_{21}P + \Theta_{12} \\ &= \Theta_{11}P + \Theta_{12}. \end{aligned}$$

Consequently,

$$\underline{G}_0 = [(\Theta_{11}P + \Theta_{12})\Gamma^{-1}]^\top = G_0^\top,$$

and hence

$$\mathbf{D}(\Theta) \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w} = \begin{bmatrix} G_0^\top \\ I \end{bmatrix} \underline{\Gamma}w.$$

By Theorem 2, G_0 and $G_0^\top$ have the same induced L_2 -gain. Hence, for any ρ ,

$$\begin{aligned} \|p_T G_0 \tilde{w}\|_{L_{[a,b]}} &\leq \rho \|p_T \tilde{w}\|_{L_{[a,b]}} \quad \text{for all } \tilde{w} \text{ and } T \geq 0 \\ \iff \|p_T G_0^\top \tilde{w}\|_{L_{[a,b]}} &\leq \rho \|p_T \tilde{w}\|_{L_{[a,b]}} \quad \text{for all } \tilde{w} \text{ and } T \geq 0. \end{aligned}$$

Since $\underline{\Gamma}$ is causally boundedly invertible, the same arguments used for 1) $\implies$ 2) and 2) $\implies$ 1), with G_0 , Γ , w , and $\mathcal{I}$ replaced by $G_0^\top$, $\underline{\Gamma}$, $\underline{w}$, and $\underline{\mathcal{I}}$, respectively, establish 1) $\iff$ 3).

Realization of G_0 . Let $\mathbf{x}$ collect the plant, filter, and any controller states, so that the controller is already included in the primal operators. Partition the filtered realization as

$$\begin{bmatrix} \mathcal{T}\dot{\mathbf{x}} \\ \tilde{z} \\ \tilde{w} \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C}_z & \mathcal{D}_z \\ \mathcal{C}_w & \mathcal{D}_w \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ w \end{bmatrix}.$$

If $\mathcal{D}_w$ is invertible, eliminating w gives the realization

$$G_0 \sim \{\mathcal{T}, \mathcal{A} - \mathcal{B}\mathcal{D}_w^{-1}\mathcal{C}_w, \mathcal{B}\mathcal{D}_w^{-1}, \mathcal{C}_z - \mathcal{D}_z\mathcal{D}_w^{-1}\mathcal{C}_w, \mathcal{D}_z\mathcal{D}_w^{-1}\}, \quad (\text{A.1})$$

with input $\tilde{w}$ and output $\tilde{z}$.

Appendix A.3. Proof of Corollary 4

The last row of the primal component block gives the PI coordinate map

$$\mathcal{M}_\Gamma := \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ \mathcal{D}_{21}\mathcal{C}_G & \mathcal{C}_2 & \mathcal{D}_{21}\mathcal{D}_G + \mathcal{D}_{22} \end{bmatrix}, \quad \mathcal{M}_{\Gamma^{-1}} := \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ -*\mathcal{D}_{21}\mathcal{C}_G & -*\mathcal{C}_2 & * \end{bmatrix}.$$

Both maps belong to Π_4 , and direct block multiplication verifies that they are inverses. In particular,

$$\begin{bmatrix} \mathbf{x}_G \\ \mathbf{x}_\Theta \\ \tilde{w} \end{bmatrix} = \mathcal{M}_\Gamma \begin{bmatrix} \mathbf{x}_G \\ \mathbf{x}_\Theta \\ w \end{bmatrix}, \quad \begin{bmatrix} \mathbf{x}_G \\ \mathbf{x}_\Theta \\ w \end{bmatrix} = \mathcal{M}_{\Gamma^{-1}} \begin{bmatrix} \mathbf{x}_G \\ \mathbf{x}_\Theta \\ \tilde{w} \end{bmatrix}.$$

Thus $\mathcal{M}_\Gamma$ is the PI congruence on the augmented state–input coordinates. The causal map $\Gamma : w \mapsto \tilde{w}$ in Theorem 3 is its induced input–output map. The construction in the proof of Theorem 3 shows that the corresponding coordinate map from the dual component block is then invertible as well.

The primal and dual LPIs are the KYP inequalities for their respective component blocks. Integration with the nonnegative storage gives statements 2 and 3 of Theorem 3, respectively. Since these are equivalent, either LPI yields the strict primal hard IQC (3). Theorem 1 then gives robust stability and robust performance.

Appendix A.4. Proof of Corollary 1

Since $\mathcal{P} \succ 0$, define $\mathcal{K}_G$ and $\mathcal{K}_\Theta$ by

$$\mathcal{Z}^* = [\mathcal{K}_G \quad \mathcal{K}_\Theta] \mathcal{P}.$$

The feedback-dependent rows of the dual component block give

$$\underline{\mathcal{A}} := \underline{\mathcal{A}}_0 + \begin{bmatrix} \mathcal{K}_G^* \\ \mathcal{K}_\Theta^* \end{bmatrix} \begin{bmatrix} \mathcal{B}_u^* & 0 \end{bmatrix}, \quad \underline{\mathcal{B}} := \underline{\mathcal{B}}_0 + \begin{bmatrix} \mathcal{K}_G^* \\ \mathcal{K}_\Theta^* \end{bmatrix} \mathcal{D}_{zu}^*.$$

Then (13) is the dual KYP inequality for the closed-loop filtered system $\mathbf{D}(\Theta) \begin{bmatrix} (K \star G)^\top \\ I \end{bmatrix}$. From the last row of the primal component block, its PI congruence transformation and inverse are

$$\mathcal{M}_\Gamma := \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ \mathcal{D}_{21}(\mathcal{C}_G + \mathcal{D}_{zu}\mathcal{K}_G) & \mathcal{C}_2 + \mathcal{D}_{21}\mathcal{D}_{zu}\mathcal{K}_\Theta & \mathcal{D}_{21}\mathcal{D}_G + \mathcal{D}_{22} \end{bmatrix},$$

$$\mathcal{M}_{\Gamma^{-1}} := \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ -*\mathcal{D}_{21}(\mathcal{C}_G + \mathcal{D}_{zu}\mathcal{K}_G) & -*(\mathcal{C}_2 + \mathcal{D}_{21}\mathcal{D}_{zu}\mathcal{K}_\Theta) & * \end{bmatrix}.$$

Equivalently, the primal filtered-input row is

$$\begin{bmatrix} \mathbf{x}_G \\ \mathbf{x}_\Theta \\ \tilde{w} \end{bmatrix} = \mathcal{M}_\Gamma \begin{bmatrix} \mathbf{x}_G \\ \mathbf{x}_\Theta \\ w \end{bmatrix}.$$

The last row of the dual component block gives the differently structured dual map. By the primal–dual invertibility equivalence in Theorem 3,

$$\underline{*} := (\underline{\mathcal{D}}_{21}\mathcal{D}_G^* + \underline{\mathcal{D}}_{22})^{-1} \in \Pi_4,$$

and

$$\mathcal{M}_{\underline{\Gamma}} := \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ \underline{\mathcal{D}}_{21}\mathcal{B}_G^* & \underline{\mathcal{C}}_2 & \underline{\mathcal{D}}_{21}\mathcal{D}_G^* + \underline{\mathcal{D}}_{22} \end{bmatrix}, \quad \mathcal{M}_{\underline{\Gamma}^{-1}} := \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ -*\underline{\mathcal{D}}_{21}\mathcal{B}_G^* & -*\underline{\mathcal{C}}_2 & \underline{*} \end{bmatrix}.$$

That row acts as

$$\begin{bmatrix} \mathbf{x}_G \\ \mathbf{x}_\Theta \\ \underline{\tilde{w}} \end{bmatrix} = \mathcal{M}_{\underline{\Gamma}} \begin{bmatrix} \mathbf{x}_G \\ \mathbf{x}_\Theta \\ \underline{w} \end{bmatrix}.$$

Unlike $\mathcal{M}_\Gamma$, the dual map has no explicit controller-gain blocks. The gains enter the dual state and input operators above. The analysis map is recovered by setting $\mathcal{K}_G = \mathcal{K}_\Theta = 0$. These PI matrices act on the augmented state–input coordinates and induce the causal maps Γ and $\underline{\Gamma}$ used in Theorem 3. They are not those input–output maps themselves. All four coordinate maps lie in Π_4 . Hence the dual KYP inequality gives statement 3 of Theorem 3, which is equivalent to the strict primal hard IQC in statement 2. Therefore Theorem 1 proves robust stability and robust performance of $P_K \star \Delta$.

Appendix A.5. Proof of Lemma 1

Suppose that the dual IQC

$$\left\langle p_T \begin{bmatrix} P_\Delta^\top w_\Delta \\ w_\Delta \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_\Delta} \begin{bmatrix} P^\top w_\Delta \\ w_\Delta \end{bmatrix} \right\rangle_{L[a,b]}$$

is parametrized according to (15). We need to show that Δ satisfies the primal IQC. First, observe the dual IQC can be factorized as

$$\mathcal{V}_\Delta = \frac{1}{2} \begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}^* \begin{bmatrix} \mathcal{S} & I \\ \mathcal{S} & -I \end{bmatrix}^* \begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix} \underbrace{\begin{bmatrix} \mathcal{S} & I \\ \mathcal{S} & -I \end{bmatrix}}_{\Theta_1} \underbrace{\begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}}_{\Theta_2} \quad (\text{A.2})$$

Then, we use the multiplicative and involutive property of the dual operation such that $\mathbf{D}(\Theta_1 \Theta_2) = \mathbf{D}(\Theta_1) \mathbf{D}(\Theta_2)$ for any two invertible Θ_1 and Θ_2 and $\mathbf{D}(\mathbf{D}(\Theta)) = \Theta$ to recover the primal IQC. Since $\underline{\mathcal{S}}_0 \succeq 0$, $\underline{\mathcal{S}}$ is closed under inversion with $\underline{\mathcal{S}}_0^{-1} \succeq 0$. Applying these properties to the two factors of $\underline{\Theta}$ in (A.2) gives

$$\begin{aligned} \Theta &= \mathbf{D}(\underline{\Theta}) = \mathbf{D}\left(\frac{1}{\sqrt{2}} \begin{bmatrix} I & \underline{\mathcal{S}} \\ I & -\underline{\mathcal{S}} \end{bmatrix}\right) \mathbf{D}\left(\begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}\right) \\ &= \frac{1}{\sqrt{2}(\beta - \alpha)} \begin{bmatrix} -\underline{\mathcal{S}}^{-1} & -I \\ -\underline{\mathcal{S}}^{-1} & I \end{bmatrix} \begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}. \end{aligned}$$

We omit $\frac{1}{(\beta - \alpha)} > 0$ therefore,

$$\mathcal{V}_\Delta = \begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}^* \begin{bmatrix} 0 & \underline{\mathcal{S}}^{-1} \\ \underline{\mathcal{S}}^{-1} & 0 \end{bmatrix} \begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}$$

For completion we use the proof in [17] Then, for all $s \in [a, b]$ and $t > 0$, we have that

$$(\beta z_\Delta(t, s) - \phi(z_\Delta(t, s)))(\phi(z_\Delta(t, s)) - \alpha z_\Delta(t, s)) \geq 0,$$

and hence

$$(\beta z_\Delta(t, s) - \phi(z_\Delta(t, s))) \underline{\mathcal{S}}_0^{-1}(s) (\phi(z_\Delta(t, s)) - \alpha z_\Delta(t, s)) \geq 0.$$

Therefore, Δ satisfies the IQC defined by $\underline{\Psi}_\Delta = I$ and (15).